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METHODS OF PREPARING COHN POOL CONCENTRATE FROM BLOOD PLASMA THROUGH ULTRAFILTRATION

Abstract

The present invention provides a method of fractionating human plasma, in some embodiments, using the Cohn fractionation procedure. The improvement comprises the use of physiologically active concentrated plasma as the starting material for the fractionation procedure.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention resides in the field of plasma fractionation to separate therapeutically active plasma proteins from plasma.

BACKGROUND OF THE INVENTION

[0002] The past decade has seen a steady increase in the clinical utilization of plasma protein-based therapeutics. For example, as awareness of primary immunodeficiency disease (PID) has increased, the effective clinical use of intravenous immunoglobulin (IVIG) has increased across the patient population affected by PID. IVIG is increasingly being utilized for off-label indications as well for conditions such as chronic inflammatory demyelinating polyneuropathy (CIPD).

[0003] In 2019, the blood plasma product market was forecast to grow at a CAGR of 6.8% to reach $28.5 B in 2023 from $20.5 B in 2018. The global annual fractionation capacity was about 70.7 million liters in 2016.

[0004] The process of plasma fractionation is infrastructure/facility intensive and highly regulated. To meet the increasing demand for plasma-derived therapeutics, either existing infrastructure must be capable of meeting this demand, or the infrastructure must be modified or increased with the latter two of these options requiring significant capital outlay, a potential interruption to the production line and potential regulatory review and certification of the modified or new facilities. Thus, an option for enhancing the efficiency of existing infrastructure without its substantial modification is highly attractive.

[0005] A method of reducing the volume of liquid plasma input into the fractionation process could achieve the dual benefits of maximizing the productivity of existing fractionation infrastructure as well as reduce the level of needed resources during plasma fractionation. A method apparently unexplored until the present invention is one in which a reduced volume of liquid is processed to produce the same amount of plasma-derived protein product as would be produced from a higher volume of liquid, e.g., Fresh-Frozen Plasma (FFP).

[0006] Accordingly, until the invention described herein, it has not been apparent that the proteins in the various fractions (e.g., cold ethanol fractions) could be recovered by fractionating a concentrated plasma Cohn Pool in amounts sufficiently meaningful to make the expense of concentrating and fractionating the concentrated physiologically active plasma worthwhile. Additionally, it was not known whether the concentrated plasma Cohn Pool would act similarly to fresh frozen plasma in Cohn Fractionation (or a known modification thereof). The inventors have discovered that a fractionation route originating with a concentrated plasma Cohn Pool is indeed feasible. The concentrated Cohn Pool is a component of an economically viable fractionation process, e.g., Cohn Fractionation or Kistler-Nitschman Fractionation, or other method (e.g., Gerlough, Hink, and Mulford methods) commencing with a concentrated plasma Cohn Pool. See, e.g., Kistler et al., *Vox. Sang.* (1962); 7(4), pp. 414-424; Graham, et al. Subcellular Fractionation, a Practical Approach. Oxford University Press. 1997.

BRIEF SUMMARY OF THE INVENTION

[0007] Given the increasingly broad use of therapeutic plasma-derived blood protein compositions, such as immune globulin compositions, albumin, protease inhibitors, blood coagulation factors, coagulation factor inhibitors, and proteins of the complement system, ensuring adequate, economical, environmentally friendly, and sustainable access to efficacious and safe plasma-derived blood protein compositions is of paramount importance.

[0008] It has now been discovered that modifying a standard plasma protein fractionation process by inputting into the process a concentrated plasma Cohn Pool leads to process efficiencies consequent to reducing the volume of liquid to be fractionated and, in some embodiments, surprisingly, essentially proportional to the increase in concentration of the plasma input introduced into the process. The improved method results in increased utilization of processing equipment and an increase in the throughput, which, in some embodiments, is roughly proportional to the concentration factor and can, therefore, be significant. In various embodiments, the invention provides an improved plasma fractionation method having one or more of these improved properties. Also provided are plasma protein products prepared using the improved procedure.

[0009] With the current invention, it has been discovered that a concentrated plasma Cohn Pool is an efficacious starting material for preparing protein therapeutic agents by fractionating the concentrated Cohn Pool. In various embodiments, the proteins typically found in the various Cohn fractions downstream from the concentrated plasma Cohn Pool are found in these fractions in yields and purity comparable to those in which they found in corresponding fractions in a process starting with a (non-concentrated) plasma feedstock, e.g., cryo poor plasma, plasma following one or more absorptive steps, recovered plasma, plasma from plasmapheresis, frozen plasma, thawed plasma and the like.

[0010] An exemplary method of the invention includes: (a) submitting a concentrated plasma Cohn Pool to one or more plasma fractionation processes (e.g., cold ethanol fractionation). In an exemplary embodiment, the invention provides, prior to (a), (b) preparing a concentrated plasma Cohn Pool.

[0011] Thus, in various embodiments, the invention provides an improved process for fractionating plasma. The improvement comprises initiating the plasma fractionation process with a concentrated plasma Cohn Pool. In various embodiments, the improvement further comprises concentrating a plasma input prior to submitting the concentrated input to a first alcohol fractionation step. An exemplary plasma input is concentrated cryo poor plasma.

[0012] In various embodiments, the reduction in plasma input volume and concomitant increase in plasma protein concentration(s) results in a plasma fractionation method that is, to a surprising degree characterized by the reduction in volume/increase in plasma protein concentration(s).

[0013] In various embodiments, the invention provides an improved plasma fractionation process proceeding to completion of a selected step within a determined amount of time. The improvement to the plasma fractionation process comprises completing the selected final step in a time reduced relative to a plasma fractionation process otherwise identical with the exception that the plasma input into the first alcohol fractionation step in the method of the invention is concentrated relative to the plasma input into the otherwise identical process. An exemplary plasma input is concentrated cryo poor plasma, thus, concentrated cryo poor plasma is the concentrated Cohn Pool in this embodiment.

[0014] For example, a fractionation process initiated with a Cohn Pool concentrated by about 10%, about 20% or about 30% relative to a standard plasma fractionation input results in the use of about 10%, about 20% or about 30% less reagents than a comparable process beginning with an unconcentrated input. Similarly, a fractionation process initiated with a Cohn Pool concentrated by about 10%, about 20% or about 30% relative to a standard plasma fractionation input results in the expenditure of about 10%, about 20% or about 30% less time from start to completion than a comparable process beginning with an unconcentrated input. This result was not expected as the mass of the plasma proteins in the concentrated input is essentially unchanged between the concentrated and unconcentrated inputs.

[0015] In an exemplary embodiment, the invention provides an improved method of preparing a protein fraction from plasma, wherein the fraction is enriched in a plasma protein product selected from a coagulation factor (e.g., factor V, VII, VIII, IX, X, XI, XII, and XIII), the prothrombin complex, Von Willebrand factor, factor VIII/Von Willebrand factor, fibrin, fibrinogen, thrombin, polyvalent and hyperimmune (such as anti-RhO, anti-hepatitis B, anti-rabies, or anti-tetanus) immunoglobulins (IgGs), protease inhibitors (such as alpha 1-antitrypsin and C1-inhibitor), anticoagulants (such as antithrombin), C1-esterase inhibitor, Protein C, and albumin, and a combination thereof. The improvement comprises, concentrating a plasma input, preparing a concentrated Cohn Pool, prior to introducing the concentrated Cohn Pool to a first alcohol fractionation step of a plasma fractionation method. An exemplary plasma input is concentrated cryo poor plasma.

[0016] The instant improved process is applicable to any plasma fractionation process, e.g., Cohn, Gerlough, Hink, Mulford, Kistler Nitschmann, heat ethanol fractionation, Hao, etc.

[0017] The present invention also provides a plasma processing system, preferably a cGMP compliant system, which is used, inter alia, to fractionate plasma from a concentrated Cohn pool input, e.g., concentrated cryo poor plasma.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. **1** is a generalized flow diagram of an exemplary Cohn fractionation procedure.

[0019] FIG. **2** is an exemplary flow diagram for concentrating starting plasma to form the concentrated Cohn pool input into the fractionation process.

[0020] FIG. **3** illustrates an exemplary improved fractionation process of the invention with the filtration step identified and located with a red dot.

DETAILED DESCRIPTION OF THE INVENTION

I. Introduction

[0021] Making up about 55% of the total volume of whole blood, blood plasma is a whole blood component in which blood cells and other constituents of whole blood are suspended. Blood plasma further contains a mixture of over 700 proteins and additional substances that perform functions necessary for bodily health, including clotting, protein storage, and electrolytic balance, amongst others. When extracted from whole blood, blood plasma may be employed to replace bodily fluids, antibodies and clotting factors. Accordingly, blood plasma is extensively used in medical treatments.

[0022] As set forth hereinabove and in the following sections, the present invention, by starting fractionation with a concentrated plasma Cohn Pool, imparts numerous efficiencies and other advantages to the fractionation process.

[0023] Reference will now be made in detail to implementation of exemplary embodiments of the present disclosure as illustrated in the accompanying drawings. The same reference indicators will be used throughout the drawings and the following detailed description to refer to the same or like parts. Those of ordinary skill in the art will understand that the following detailed description is illustrative only and is not intended to be in any way limiting. Other embodiments of the present disclosure will readily suggest themselves to such skilled persons having benefit of this disclosure.

[0024] In the interest of clarity, not all of the routine features of the implementations described herein are shown and described. It will be appreciated that, in the development of any such actual implementation, numerous implementation-specific decisions are made in order to achieve the plasma product producer's specific goals, such as compliance with application-and business-related constraints, and that these specific goals will vary from one implementation to another and from one plasma product producer to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine undertaking of engineering for those of ordinary skill in the art having the benefit of this disclosure.

[0025] Many modifications and variations of the exemplary embodiments set forth in this disclosure can be made without departing from the spirit and scope of the exemplary embodiments, as will be apparent to those skilled in the art. The specific exemplary embodiments described herein are offered by way of example only, and the disclosure is to be limited only by the terms of the appended claims, along with the full scope of equivalents to which such claims are entitled.

II. Abbreviations and Definitions

[0026] Unless defined otherwise, all technical and scientific terms used herein generally have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Generally, the nomenclature used herein and the laboratory procedures in organic chemistry, pharmaceutical formulation, and medical imaging are those well-known and commonly employed in the art.

b. Definitions

[0027] The articles “a” and “an” are used herein to refer to one or to more than one (i.e., to at least one) of the grammatical object of the article. By way of example, “a protein” means one protein or more than one protein.

[0028] The “Cohn Process”, and “Cohn Fractionation” are used interchangeably herein and as generally understood, refer to a method of separating human plasma through a series of steps, including ethanol precipitation at differing concentrations, changes in pH, changes in temperature, changes in ionic strength, which lead to fractions enriched in certain plasma proteins. See, for example U.S. Pat. No, 2,390,074. FIG. **1** provides an exemplary flow diagram for the Cohn Process. As used herein, the terms “Cohn Process” and “Cohn Fractionation” also refers to the many variations and improvements on this pioneering process, e.g., Kistler-Nitschmann Process (Kistler et al. (1952), *Vox Sang*, 7, 414-424). Other processes of use in the methods of the invention include the method of isolating IgG set forth in U.S. Pat. No. 8,940,877

[0029] “Plasma” is the fluid that remains after blood has been centrifuged (for example) to remove cellular materials such as red blood cells, white blood cells and platelets. Plasma is generally yellow-colored and clear to opaque. Blood that is donated and processed to separate the plasma from the other certain blood components, and not frozen is referred to as “never-frozen” plasma. Plasma that is frozen within 8 hours to temperatures, described herein, is referred to herein as “fresh frozen plasma” (“FFP”). It contains the dissolved constituents of the blood such as proteins (6-8%; e.g., serum albumins, globulins, fibrinogen, etc.), glucose, clotting factors (clotting proteins), electrolytes (Na.sup.+, Ca.sup.2+, Mg.sup.2+, HCO.sub.3.sup.−, Cl.sup.−, etc.), hormones, etc. Whole blood (WB) plasma is plasma isolated from whole blood with no added agents except anticoagulant(s). Citrate phosphate dextrose (CPD) plasma, as the name indicates, contains citrate, sodium phosphate and a sugar, usually dextrose, which are added as anticoagulants.

[0030] “Recovered plasma” refers to plasma separated no later than 5 days after the expiration date of the Whole Blood and is stored at 1 to 6° C. The profile of plasma proteins in Liquid Plasma is poorly characterized. Levels and activation state of coagulation proteins in Liquid Plasma are dependent upon and change with time in contact with cells, as well as the conditions and duration of storage. This component serves as a source of plasma proteins. Levels and activation state of coagulation proteins are variable and change over time.

[0031] “Thawed plasma” refers to plasma derived from Source, FFP or FP24, prepared using aseptic techniques (closed system), thawed at from about 10 to about 37° C., and maintained at from about 1 to about 6° C. for up to about 4 days after the initial 24-hour post-thaw period has elapsed. Thawed plasma contains stable coagulation factors such as Factor II and fibrinogen in concentrations similar to those of FFP, but variably reduced amounts of other factors. An exemplary thawed plasma is a component of the first fractionation step where the plasma (source, Recovered, FF etc) is removed from plastic containers and thawed in a jacketed vessel (with an exchange fluid at the temperature up 37° C.)

[0032] “Fresh frozen plasma” (“FFP”) refers to plasma prepared from a whole blood or apheresis collection and frozen at about −18° C. or colder within the time frame as specified in the directions for use for the relevant blood collection, processing, and storage system (e.g., frozen within eight hours of draw). On average, units contain 200 to 250 mL, but apheresis derived units may contain as much as 400 to 600 mL. FFP contains plasma proteins including all coagulation factors. FFP

contains high levels of the labile coagulation Factors V and VIII.
[0033] As used herein, a "Factor" followed by a Roman Numeral refers to a series of plasma proteins which are related through a complex cascade of enzyme-catalyzed reactions involving the sequential cleavage of large protein molecules to produce peptides, each of which converts an inactive zymogen precursor into an active enzyme leading to the formation of a fibrin clot. They include: Factor I (fibrinogen), Factor II (prothrombin), Factor III (tissue thromboplastin), Factor IV (calcium), Factor V (proaccelerin), Factor VI (no longer considered active in hemostasis), Factor VII (proconvertin), Factor VIII (antihemophilic factor), Factor IX (plasma thromboplastin component; Christmas factor), Factor X (Stuart factor), Factor XI (plasma thromboplastin antecedent), Factor XII (hageman factor), and Factor XIII (fibrin stabilizing factor).
[0034] A "plasma protein" includes, for example, coagulation factors (such as factor V, VII, VIII, IX, X, XI, XII, and XIII), the prothrombin complex, Von Willebrand factor, factor VIII/Von Willebrand factor, fibrin, fibrinogen, thrombin, polyvalent and hyperimmune (such as anti-RhO, anti-hepatitis B, anti-rabies, or anti-tetanus) immunoglobulins (IgGs), protease inhibitors (such as alpha 1-antitrypsin and C1-inhibitor), anticoagulants (such as antithrombin), C1-esterase inhibitor, Protein C, and albumin, and a combination thereof.
[0035] As used herein, the term concentrated plasma "Cohn Pool" refers to a plasma pool, which, in the method of the invention, is concentrated and subsequently undergoes a plasma fractionation process (e.g., Cryo poor, Coagulation Factors poor, Inhibitors poor plasma). An exemplary concentrated plasma Cohn Pool is a physiologically active plasma concentrated by from about 10% to about 30% from its original volume (e.g., volume upon collection from a donor or donor pool, receipt by a fractionation facility, etc.), and which includes proteins that have not been damaged to such an extent to lose substantially all of their physiological activity.
[0036] In an exemplary embodiment, the concentration process results in essentially no diminution in the activity of a selected plasma protein subsequently isolated (or enriched) by fractionation of the concentrated Cohn Pool, e.g., IgG, A1PI, a Factor, etc. In various embodiments, the activity of a selected plasma protein fractionated from the physiologically active concentrated Cohn Pool is not less than about 99%, not less than abut 90%, not less than about 85% or not less than about 80% of the activity of the selected plasma protein in plasma feedstock before its concentration. In various embodiments, the selected plasma protein is polyvalent or hyperimmune IgG.
[0037] A "disease" is a state of health of an animal wherein the animal cannot maintain homeostasis, and wherein if the disease is not ameliorated then the animal's health continues to deteriorate. In various embodiments, one or more proteins from the fractionated physiologically active concentrated Cohn Pool are used to treat one or more disease.

III. Embodiments

A. Compositions and Devices

[0038] Embodiments of the present disclosure are directed to methods of concentrating a physiologically active plasma feedstock to form a concentrated Cohn Pool and subsequently fractionating the resulting physiologically active concentrated Cohn Pool using an art-recognized fractionation process. An exemplary fractionation process includes as its first step an alcohol fractionation step. Also provided are plasma protein preparations prepared by a fractionation process commencing with the concentrated Cohn Pool.
[0039] In an exemplary embodiment, the invention provides a physiologically active concentrated plasma. The biological and physiological activity of the concentrated plasma is essentially non-degraded when compared to the starting plasma from which the concentrated plasma was derived. By essentially non-degraded is meant that for any selected plasma protein, its activity in the concentrated plasma is not less than about 80%, not less than about 85%, not less than about 90%, not less than about 95%, or not less than about 99% of its activity in the starting plasma. In exemplary embodiments, this is true of at least two selected plasma proteins, at least five selected proteins or at least ten selected proteins. In an exemplary embodiment, the overall plasma protein

activity of the concentrated plasma is essentially non-degraded when compared to the starting plasma.

[0040] In an exemplary embodiment, the concentrated plasma of the invention includes essentially no greater fraction of plasma protein aggregates than were present in the starting plasma. By essentially no greater fraction of plasma protein aggregates is meant, not more than about 2%, not more than about 5%, not more than about 10%, not more than about 15% or not more than about 20% more aggregates on a wt % basis. The wt % basis is calculated from total protein weight in the concentrated plasma pool and starting plasma, i.e., not more than about X % of the plasma protein in the concentrated plasma pool is present as aggregates.

[0041] In an exemplary embodiment, the plasma protein fraction isolated according to the method of the invention has characteristics substantially identical to those of the same fractions isolated in the same manner from plasma that has not been concentrated (e.g., frozen plasma) using art-recognized methods. In various embodiments, the characteristics of the plasma protein fraction vary from those of the same protein fractions isolated in the same manner from non-concentrated plasma using art-recognized methods. In a preferred embodiment, the characteristic(s) varying between the two plasma protein fractions correspond(s) to one or more parameter of regulatory relevance, necessary for marketing approval of a therapeutic plasma protein, and the characteristic varies within a range of such one or more parameter by an amount considered insignificant with respect to relevant regulatory requirements for that fraction, i.e., a pharmaceutical formulation incorporating a plasma fraction or a protein isolated from a plasma fraction downstream from the concentrated plasma does not require new regulatory consideration or marketing approval. In various embodiments, the plasma fraction or protein isolated downstream from the concentrated plasma is essentially identical to the corresponding plasma fraction or protein isolated from non-concentrated plasma.

[0042] In an exemplary embodiment, the concentrated plasma pool is a starting input for an improved fractionation process. In various embodiments, the concentrated plasma pool improves the process by facilitating and/or promoting one or more of: (i) decreasing fractionation time, (ii) decreasing fractionating material outlay, (iii) decreasing waste, the use of pollutants, e.g., of VOCs, and (iv) increasing throughput with existing fractionating plant infrastructure. In an exemplary embodiment, these results are achieved with essentially no reduction in yield of a selected plasma protein fraction. By essentially no reduction in yield in this instance is meant, when compared to an identical fractionation process commencing with a non-concentrated plasma input, the overall yield of plasma protein is not less than about 2%, not less than about 5%, not less than about 10%, not less than about 15% or not less than about 20% of the overall yield of plasma protein from the process commencing with the non-concentrated input.

[0043] In various embodiments, the processing of the physiologically active concentrated plasma is conducted with the addition of one or more component used in plasma fractionation, e.g., alcohol, acid, base. In an exemplary embodiment, the physiologically active concentrated plasma is maintained or passes through a component of a fractionation system, and is incorporated into a process using such a system. In an exemplary embodiment, the fractionation system is a Cohn fractionation system, or a known modification of this system.

[0044] In an exemplary embodiment, the invention provides one or more plasma protein fractions, product(s) of a plasma fractionation process commencing with the physiologically active concentrated plasma. In an exemplary embodiment, the plasma protein fraction is a Cohn fraction as this term is understood in the art.

[0045] In various embodiments, the invention provides one, two, three, four, five or more unique plasma fraction composition(s) downstream from a physiologically active concentrated plasma input. In various embodiments, the composition is Fraction I paste and comprises fibrinogen, or Fraction I supernatant. In various embodiments, the composition is Fraction II+III (or Fr. I+II+III) paste and comprises IgG, or Fraction II+III (or Fraction I+II+III) supernatant. In some embodiments, the composition is Fraction IV-1 paste and comprises A1PI and/or AT-III, or Fraction IV-1 supernatant. In an exemplary embodiment, the plasma fraction composition is Fraction IV-4 paste and/or Fraction IV-4 supernatant. In various embodiments, the plasma fraction composition is Fraction V paste and comprises albumin, or Fraction V supernatant. In an exemplary embodiment, the fraction or fractions is/are one or more Cohn fraction.

[0046] In an exemplary embodiment, the invention provides a preparation of a protease inhibitor prepared by a method of the invention. In various embodiments, the protease inhibitor is selected from alpha 1-antitrypsin, C1-inhibitor, etc.) and a combination thereof.

[0047] In an exemplary embodiment, the invention provides a preparation of albumin prepared by a method of the invention.

[0048] In an exemplary embodiment, the method provides an aqueous albumin solution containing at least about 5% or at least about 25% by volume of albumin and is suitable for intravenous injection in a human subject, which solution remains stable without precipitation of the albumin after exposure to a temperature of about 45° C. for a period of one month. In an exemplary embodiment, this solution is isolated by fractionation of physiologically active concentrated human plasma of the invention.

[0049] In an exemplary embodiment, the invention provides a preparation of IgG isolated from the physiologically active concentrated human plasma. The preparation comprises the IgG in an amount of not less than 60%, 65%, 70%, 75%, 80%, 85%, 90%, or 95% of the amount found in an identical preparation in which IgG is isolated from non-concentrated plasma (e.g., fresh frozen plasma). In various embodiments, the activity of the IgG is not less than 60%, 65%, 70%, 75%, 80%, 85%, 90%, or 95% of the activity of the IgG isolated from non-concentrated plasma (e.g., fresh frozen plasma). The IgG can be polyvalent or hyperimmune IgG.

[0050] In an exemplary embodiment, the invention provides a plasma protein isolated from Fraction IV-1 of the fractionated physiologically active concentrated human plasma selected from A1PI, AT-III and a combination thereof. In an exemplary embodiment, the plasma protein is isolated in a yield of not less than about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, about 90%, or about 95% of the yield in which this protein is isolated from non-concentrated plasma (e.g., fresh frozen plasma). In various embodiments, the protein isolated from the physiologically active concentrated human plasma in Fraction IV-1 has an activity of not less than about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, about 90%, or about 95% of the activity of the protein isolated from non-concentrated plasma (e.g., fresh frozen plasma).

[0051] In some embodiments, the invention provides a method wherein albumin isolated from Fraction V of the physiologically active concentrated human plasma is isolated in a yield of not less than about 80% of the yield in which this protein is isolated from a non-concentrated plasma input, e.g., fresh frozen plasma. In various embodiments, the albumin isolated from the physiologically active concentrated human plasma has an activity of not less than about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, about 90%, or about 95% of the activity of albumin isolated from non-concentrated plasma (e.g., fresh frozen plasma).

[0052] In various embodiments, the invention provides a pharmaceutical formulation comprising one of the plasma protein fractions produced by a method of the invention, or a protein component of one or more such fraction further purified from such fraction. Various pharmaceutical formulations also include a pharmaceutically acceptable vehicle in which the plasma fraction or proteins in/from the fraction (or downstream where further purified) arc formulated.

[0053] In various embodiments, the invention provides a pharmaceutical formulation of the invention packaged in a device for administering the pharmaceutical formulation to a subject in need of such administration, e.g., a syringe, infusion bag, and the like. In various embodiments, the device contains a unit dosage formulation of the active protein for administration to a subject in need of such administration. In an exemplary embodiment, the unit dosage is an art-recognized unit dosage for a subject.

B. Methods
[0054] In various embodiments, the present invention provides a novel method of plasma fractionation commencing with a physiologically active concentrated plasma of the invention as the input. An exemplary method of the invention includes providing a physiologically active concentrated plasma solution prepared by, in an exemplary embodiment, ultrafiltration; and submitting the physiologically active concentrated plasma to one or more fractionation process. is Cohn fractionation (FIG. **1**), and variations thereof.
[0055] In an exemplary embodiment, the starting plasma of use in the methods of the present invention is concentrated after pooling (Cryo-poor, Coagulation Factors-poor, Inhibitors-poor plasma pool). Concentration of the Cohn pool can be achieved by a number of techniques including, but not limited to tangential flow filtration, ultrafiltration and a combination thereof. For example, methods of concentrating the starting plasma include: (a) pre-filtration followed by batch TFF with UF cassettes or (b) pre-filtration followed by single-pass TFF with UF cassettes or (c) UF hollow fiber system with or without pre-filtration.
[0056] FIG. **3** illustrates an exemplary improved fractionation process of the invention with the filtration step identified and located with a red dot.
[0057] The physiologically active concentrated Cohn Pool is, in one embodiment, concentrated through ultrafiltration. The ultrafiltration can be performed in any useful format (i.e., order of addition, temperature, dilution, etc.).
[0058] Proteins potentially undergo physical degradation by a number of mechanisms (e.g., clipping, oxidation, unfolding, aggregation, insoluble particulate formation). Many proteins are structurally unstable in solution and are susceptible to conformational changes due to various stresses encountered during purification, processing and storage. These stresses include temperature shift, exposure to pH changes and extreme pH, shear stress, surface adsorption/interface stress, and so on.
[0059] As will be appreciated by those of skill in the art, any of these modes of Cohn Pool concentration can be performed singly or in any combination or order.
In various embodiments, the physiologically active concentrated Cohn Pool is composed of at least about 65 g/L plasma protein.
[0060] In various embodiments, following cryoprecipitation, the plasma is separated into cryoprecipitate and cryosupernatant. The cryosupernatant or cryosupernatant after adsorption is optionally submitted to further fractionation steps. The separation may be accomplished in any useful fashion, such as, without limitation, centrifugation, filtration or a combination thereof.
[0061] In those embodiments in which cooling of the physiologically active concentrated plasma is desired, any useful means of cooling can be utilized. In various embodiments, a vessel or line containing the concentrated plasma is jacketed with a cooling device. In exemplary embodiments, the cooling and/or plasma solution is retained in a vessel, e.g., a jacketed vessel, and, in some embodiments, the plasma solution is cooled during inline flow (“radiator method”).
[0062] In some embodiments, the physiological concentrated plasma contains albumin in an amount from about 3.5 to about 5.5 g/dL. In various embodiments, the albumin concentration of the physiologically active concentrated plasma is from about 40% to about 70%, e.g., from about 50% to about 60% of the total plasma protein content of the physiologically active concentrated plasma.
[0063] In various embodiments, the albumin in the physiologically active concentrated plasma retains at least about 80%, 85%, 90%, or at least about 95% of the activity on a per unit basis of albumin in plasma.
[0064] In some embodiments, the physiological concentrated plasma contains A1PI in an amount from about 50-300 mg/dL, e.g., from about 100 to about 200 mg/dL.
[0065] In various embodiments, the A1PI in the physiologically active concentrated plasma retains at least about 80%, 85%, 90%, or at least about 95% of the activity on a per unit basis of A1PI in

plasma.
[0066] In various embodiments, the physiological concentrated plasma contains IgG in an amount of from about 500 to about 1600 mg/dL, e.g., from about 700 to about 1500 mg/dL.
[0067] In various embodiments, the IgG in the physiologically active concentrated plasma retains at least about 80%, 85%, 90%, or at least about 95% of the activity on a per unit basis of IgG in plasma.
[0068] In some embodiments, the physiologically active concentrated plasma has an average particle size of about 30 microns or less. In some embodiments, the physiologically active concentrated plasma has a maximum particle size of about 100 microns or less.
[0069] In some embodiments, the physiologically active concentrated plasma includes at least 30% plasma protein by weight.
[0070] In some embodiments, the physiologically active concentrated plasma is sterile.
[0071] In an exemplary embodiment, the invention provides a method of fractionating physiologically active concentrated human plasma using the Cohn fractionation procedure, for example, that procedure set forth in U.S. Pat. No. 2,390,074, wherein the instant improvement comprises the use of physiologically active concentrated human plasma as the starting material for the fractionation procedure. FIG. **1** provides an exemplary process diagram for a method of Cohn fractionation.
[0072] Thus, for example, the physiologically active concentrated plasma is submitted to a method of fractionating proteins. An exemplary method involves precipitating a selected protein fraction from a solution containing a plurality of protein fractions. The solution is adjusted to have a pH above the iso-electric point one or more protein in the fraction desired to be precipitated. In an exemplary embodiment, the pH of the fractionation solution is lowered to bring the solution same to approximately the iso-electric point of the desired fraction to be precipitated. An exemplary method comprises bringing the ionic strength of the solution to between 0.1 and 0.2. Various methods include lowering the temperature of the solution to between approximately 0° C. and the freezing point of the solution. In some embodiments an organic precipitant for the plasma protein fraction is added to the protein solution, the amount of the precipitant added being such as to cause precipitation of the desired fraction from the protein solution the said temperature, and separating the precipitate from the solution. In a preferred embodiment, the conditions are adjusted such that substantially only the desired plasma protein fraction precipitates from the solution.
[0073] In various embodiments, there is provided a method of fractionating proteins by precipitation from a solution of physiologically active concentrated human plasma containing a plurality of protein fractions, comprises bringing the pH of the solution to approximately the iso-electric point of the desired protein fraction to be precipitated, bring the ionic strength of the solution to between 0.01 and 0.2, lowering the temperature of the solution to between approximately 0° C., and the freezing point of the solution, adding and organic precipitant for protein to the protein solution, the amount of the precipitant added, the pH, the ionic strength and the temperature being such as to cause precipitation of only the desired fraction from the protein solution, and separating the precipitate from the solution.
[0074] In various embodiments, in the method for fractionating proteins from a solution of physiologically active concentrated human plasma, the steps which comprise mixing with a solution of proteins an organic precipitant for protein, adjusting the temperature between 0 and −15° C., the amount of the precipitant from about 8% to about 40%, the pH from about 4.4 to about 7 and the ionic strength from about 0.05 to about 0.2, and separating from the resulting liquid system a protein precipitated which is insoluble therein.
[0075] In some embodiments, in the method for fractionating proteins from a solution of physiologically active concentrated human plasma, the steps which comprise mixing with a solution of proteins an organic precipitant for protein, adjusting and maintaining the temperature above the freezing point thereof but not above 0° C., the amount of the precipitant from about 10%

to about 40%, the pH from about 4.4 to about 7 and the ionic strength from about 0.05 to about 0.2, and separating from the resulting liquid system a protein precipitated which is insoluble therein.

[0076] In some embodiments, there is provided a method for fractionating proteins from physiologically active concentrated human plasma, the steps which comprise adding to a containing a mixture of proteins, both an electrolyte and an organic precipitant for protein, the electrolyte being added in an amount sufficient to bring the ionic strength from about 0.01 and 0.2, and the precipitant being added in amount such as to cause precipitation of only the desired protein fraction, adjusting and maintaining the pH of the solution from about 4.4 to about 7 and the temperature thereof from about 0 to about −15° C., thereby precipitating a protein from the resulting system.

[0077] In an exemplary embodiment, the invention provides a method of purifying and crystallizing albumin from a solution of concentrated human plasma, which comprises dissolving impure albumin in an alcohol solution containing from about 15 to about 40% alcohol, at a pH of from about 5.5 to to about 6.0, an ionic strength of from about 0.05 to about 0.5 and at a temperature of from about 0° C. to about −5° C., and maintaining the solution within the temperature range until albumin crystallizes from the mixture.

[0078] In an exemplary embodiment, in a method of fractionating substances (e.g., proteins) having differing solubilities from a solution of concentrated human plasma at a controlled temperature and hydrogen ion concentration, removing the precipitate thus formed and precipitating a plurality of successive fractions of said substances by variation in one or more of the factors.

[0079] In various embodiments, the invention provides a method of preventing denaturation of proteins by modifying reagents which would normally result in denaturation, the method comprising adding the reagents to a protein solution of concentrated human plasma by diffusion through a semi-permeable membrane.

[0080] In one embodiment, there is provided a method for fractionating proteins from a solution of physiologically active concentrated human plasma comprising contacting the physiologically active concentrated human plasma with an organic precipitant. An exemplary embodiment includes controlling one or more of the amount of the contacting precipitant, the temperature, the hydrogen ion concentration and the ionic strength of the resulting mixture, separating the resulting precipitate from the supernatant, and separating successive protein fractions by varying a plurality of said factors affecting solubility thereof.

[0081] In an exemplary embodiment, the organic precipitant is added a temperature of about 0° or less than about 0° C.

[0082] In an exemplary embodiment, the organic precipitant is an alcohol. In various embodiments, it is added a temperature of about 0° or less than about 0° C.

[0083] In an exemplary embodiment, there is provided a method of fractionating proteins from a solution of physiologically active concentrated human plasma which comprises, precipitating one or a plurality of different protein fractions from the plasma by the physiologically active concentrated human plasma with an organic precipitant (e.g., alcohol) and by varying the temperature of the mixture of the physiologically active concentrated human plasma and the organic precipitant, the temperature being progressively lowered and the organic precipitant concentration of the mixture being increased, with the precipitation of successive protein fractions. The temperature and the percentage of alcohol are correlated so that the temperature employed for the precipitation of any given protein fraction is close to but above the freezing point of the mixture at the percentage of alcohol and plasma present therein.

[0084] Exemplary organic precipitants include ethanol, acetone, dioxane and combinations thereof.

[0085] In an exemplary embodiment, IgG isolated from the physiologically active concentrated human plasma is isolated in a yield of not less than about 60%, 65%, 70%, 75%, 80%, 85%, 90%, or not less than about 95% of the yield in which this protein is isolated from fresh frozen plasma. In various embodiments, the activity of the IgG is not less than about 60%, 65%, 70%, 75%, 80%,

85%, 90%, or not less than about 95% of the activity of the IgG isolated from fresh frozen plasma.
[0086] In an exemplary embodiment, a protein isolated from Fraction IV-1 of the fractionated physiologically active concentrated human plasma selected from A1PI, AT-III and a combination thereof is isolated in a yield of not less than about 60%, 65%, 70%, 75%, 80%, 85%, 90%, or not less than about 95% of the yield in which this protein is isolated from fresh frozen plasma. In various embodiments, the protein isolated from the physiologically active concentrated human plasma in Fraction IV-1 has an activity of not less than about 60%, 65%, 70%, 75%, 80%, 85%, 90%, or not less than about 95% of the activity of the protein isolated from fresh frozen plasma.
[0087] In some embodiments, the invention provides a method wherein albumin isolated from Fraction V of the physiologically active concentrated human plasma is isolated in a yield of not less than about 80% of the yield in which this protein is isolated from fresh frozen plasma. In various embodiments, the albumin isolated from the physiologically active concentrated human plasma has an activity of not less than about 60%, 65%, 70%, 75%, 80%, 85%, 90%, or not less than about 95% of the activity of albumin isolated from fresh frozen plasma.
[0088] The methods provided herein allow for the preparation of A1PI compositions having very high levels of purity. For example, in one embodiment, at least about 95% of the total protein in an A1PI composition provided herein is A1PI. In other embodiments, at least about 96% of the protein in this composition is A1PI, or at least about 97%, 98%, 99%, 99.5%, or more of the total protein of the composition is A1PI.
[0089] Similarly, the methods provided herein allow for the preparation of A1PI compositions containing extremely low levels of contaminating agents. For example, in certain embodiments, A1PI compositions are provided that contain less than about 10 mg/L contaminant. In other embodiments, the A1PI composition will contain less than about 5 mg/L contaminant, preferably less than about 3 mg/L contaminant, most preferably less than about 2 mg/L contaminant.
[0090] In various embodiments, the A1PI in the physiologically active concentrated plasma retains at least about 80%, 85%, 90%, or at least about 95% of the activity on a per unit basis of A1PI in plasma.
[0091] In one embodiment, the present invention provides aqueous IgG compositions comprising a protein concentration of from about 150 g/L and about 250 g/L. In certain embodiments, the protein concentration of the IgG composition is from about 175 g/L and about 225 g/L, or from about 200 g/L and about 225 g/L, or any suitable concentration within these ranges, for example at or about, 150 g/L, 155 g/L, 160 g/L, 165 g/L, 170 g/L, 175 g/L, 180 g/L, 185 g/L, 190 g/L, 195 g/L, 200 g/L, 205 g/L, 210 g/L, 215 g/L, 220 g/L, 225 g/L, 230 g/L, 235 g/L, 240 g/L, 245 g/L, 250 g/L, or higher. In a preferred embodiment, the aqueous IgG composition comprises a protein concentration of at or about 200 g/L. In a particularly preferred embodiment, the aqueous IgG composition comprises a protein concentration of at or about 204 g/L.
[0092] The methods provided herein allow for the preparation of IgG compositions having very high levels of purity. For example, in one embodiment, at least about 95% of the total protein in an IgG composition provided herein will be IgG. In other embodiments, at least about 96% of the protein is IgG, or at least about 97%, 98%, 99%, 99.5%, or more of the total protein of the composition will be IgG.
[0093] Similarly, the methods provided herein allow for the preparation of IgG compositions containing extremely low levels of contaminating agents. For example, in certain embodiments, IgG compositions are provided that contain less than about 100 mg/L IgA. In other embodiments, the IgG composition will contain less than about 50 mg/L IgA, preferably less than about 35 mg/L IgA, most preferably less than about 20 mg/L IgA. In an exemplary embodiment, the IgG preparation contains less than or equal to about 0.14 mg/ml IgA.
[0094] In some embodiments, the invention provides a preparation of polyvalent and/or hyperimmune immunoglobulins (IgGs) prepared by a method of the invention. In various embodiments, the IgG is selected from anti-RhO hyperimmune immunoglobulin, anti-hepatitis B

hyperimmune immunoglobulin, anti-rabies hyperimmune immunoglobulin, anti-tetanus IgG hyperimmune immunoglobulin and a combination of any two or more thereof.
[0095] In various embodiments, the IgG in the physiologically active concentrated plasma retains at least about 80%, 85%, 90%, or at least about 95% of the activity on a per unit basis of IgG in plasma.
Cohn Pool Concentration Process
[0096] Concentration of the Cohn Pool can be achieved by a number of techniques including (a) Pre-filtration followed by batch TFF with UF cassettes or (b) pre-filtration followed by single-pass TFF with UF cassettes or (c) UF hollow fiber system with or without pre-filtration.
[0097] Apparatuses and methods for ultrafiltration are known in art. FIG. **2** provides a process flow diagram of an exemplary method/device of use to concentrate starting Cohn Pool.
[0098] An ultrafiltration membrane or ultrafiltration hollow-fiber filter membrane with a nominal molecular weight cut off (NMWCO) of about 300 kDa or less can be used in the concentration of the Cohn Pool, either in re-circulation or in single-pass configuration, and either with or without preceding pre-filtration. In another embodiment, an ultrafiltration membrane or ultrafiltration hollow-fiber filter membrane with a nominal molecular weight cut off (NMWCO) of about 200 kDa or less can be used in the concentration of the Cohn Pool, either in re-circulation or in single-pass configuration, and either with or without preceding pre-filtration. In some embodiment, an ultrafiltration membrane or ultrafiltration hollow-fiber filter membrane with a nominal molecular weight cut off (NMWCO) of about 100 kDa or less can be used in the concentration of the Cohn Pool, either in re-circulation or in single-pass configuration, and either with or without preceding pre-filtration.
[0099] The following Examples are offered to illustrate exemplary embodiments of the invention and do not define or limit its scope.
EXAMPLES
Example 1
[0100] 4 study runs were performed until Precipitate G (PptG) and Fraction V (Fr V), respectively. The desired Cohn Pool plasma volume and protein concentration were obtained without significant loss of various plasma derived proteins from different fractionations of Cohn Pool plasma.
[0101] For each pair of study runs, i.e., Run 1 and Control 1, Run 2 and Control 2, etc. the same Cohn Pool Batch has been used as starting material. Prior to Cohn fractionation process, approx. 5-12 liters cryo-poor plasma were concentrated in 4 test runs (Run 1, Run 2, Run 3, Run 4) using an ultrafiltration membrane while no ultrafiltration was performed in the 4 respective control runs. For the plasma concentration step, various pre-filters and various TFF filters with a nominal molecular weight cut off (NMWCO) of 100 kDa or less were employed. As a result, the starting protein concentrations of the Cohn Pool plasma in the test runs were 10%-27% higher than that in control run. The study run configuration is shown in Table.
TABLE-US-00001 TABLE Study Run Configuration Overview Description Run 1 Run 2 Run 3 Run 4 Concentration 27% 25% 17% 10% Factor Pre-Filter 1.0 μm PP/ 1.0 μm PP/ 0.65 μm GF 1.0 μm PP/ 0.45 μm Nylon 0.45 μm Nylon 0.45 μm Nylon TFF Recirculation Single Pass Recirculation Recirculation Configuration TFF MWCO 100 kDa 30 kDa 30 kDa 100 kDa TFF Type PES PES Cellulose PES Acetate
[0102] For each protein of interest comparative testing was performed in both concentrated test and non-concentrated control run, respectively. All protein contents in the upstream IgG process until Precipitate G (PptG) in the test runs are overall comparable to the control run as shown in Table-Table.
TABLE-US-00002 TABLE Protein step yields - Cohn Pool Control 1 Test 1 Control 2 Test 2 Control 3 Test 3 Control 4 Test 4 Eq. CPP L 6.0 6.0 6.0 6.31 6.0 6.2 6.0 6.0 TP by Biuret g/dL Eq 5.5 5.5 5.2 5.1 5.76 5.71 5.72 5.54 CPP IgG 5.62 5.61 5.0 5.0 6.10 6.10 5.5 5.3 Albumin 32.2 32.3 31.1 29.6 32.3 33.2 33.5 33.0 AAT 1.29 1.23 1.23 1.17 1.30 1.29 1.26 1.24 AMG g/L Eq 1.09 1.03

1.05 1.03 1.07 1.11 1.07 1.06 CPP HPT 1.03 0.98 1.00 0.98 1.06 1.09 1.08 1.06 AAG 0.72 0.69 0.70 0.68 0.72 0.72 0.73 0.70 CER 0.21 0.20 0.21 0.20 0.22 0.21 0.20 0.21 Fibrinogen 2.29 2.11 1.96 1.98 2.94 3.01 2.12 2.33 Fibrinogen μg/mL 1976 1892 1802 1949 2554 2635 1853 1895 @5% TP

TABLE-US-00003 TABLE Protein step yields - Fraction I Centrifugate Control 1 Test 1 Control 2 Test 2 Control 3 Test 3 Control 4 Test 4 Eq. CPP L 6.0 6.0 6.0 6.31 6.0 6.2 6.0 6.0 TP by Biuret g/dL Eq CPP 5.80 5.58 5.44 5.07 5.50 5.48 5.45 5.23 IgG 5.58 5.33 5.1 4.9 5.81 5.86 5.52 5.38 Albumin 32.0 30.9 30.6 29.6 32.3 32.2 32.6 32.8 IgA 1.50 1.42 1.46 1.41 1.55 1.56 1.59 1.54 IgM 0.52 0.45 0.44 0.43 0.55 0.53 0.59 0.51 C3 1.06 0.98 1.03 0.96 1.17 1.12 1.11 1.09 TRF g/L Eq CPP 2.11 2.01 2.08 2.01 1.98 2.00 2.16 2.10 AAT 1.24 1.19 1.20 1.19 1.28 1.21 1.23 1.21 AMG 1.06 1.01 1.03 1.03 1.05 1.07 1.09 1.02 HPT 1.00 0.97 0.97 0.95 1.04 1.04 1.06 1.02 AAG 0.70 0.68 0.69 0.68 0.69 0.69 0.72 0.71 CER 0.20 0.18 0.20 0.19 0.21 0.21 0.20 0.21 Fibrinogen μg/mL @5% TP 314 300 359 316 377 369 355 312

TABLE-US-00004 TABLE Protein step yield - Fraction II + III Filtrate Control 1 Test 1 Control 2 Test 2 Control 3 Test 3 Control 4 Test 4 Eq. CPP L 6.0 6.0 6.0 6.31 6.0 6.2 6.0 6.0 TP by Biuret g/dL Eq 3.60 3.23 3.51 3.25 3.49 3.41 3.43 3.70 CPP IgG 0.12 0.08 0.09 0.08 0.100 0.104 0.13 0.09 Albumin 28.4 26.0 28.0 25.4 27.6 27.3 29.9 29.4 IgA 0.119 0.070 0.030 0.057 0.037 0.067 0.238 0.085 IgM 0.010 0.007 0.010 0.007 0.010 0.008 0.010 0.009 C3 0.011 0.007 0.011 0.007 0.010 0.009 0.010 0.009 TRF g/L Eq 1.75 1.57 1.77 1.63 1.67 1.69 1.75 1.73 CPP AAT 0.59 0.52 0.57 0.58 0.55 0.52 0.74 0.60 AMG 0.15 0.13 0.14 0.16 0.18 0.15 0.15 0.12 HPT 0.87 0.78 0.82 0.79 0.91 0.93 0.89 0.90 AAG 0.61 0.54 0.58 0.56 0.61 0.60 0.61 0.61 CER 0.13 0.10 0.12 0.10 0.12 0.11 0.12 0.11

TABLE-US-00005 TABLE Protein step yields - Fraction II + III Paste Extract CUNO Filtrate Control 1 Test 1 Control 2 Test 2 Control 3 Test 3 Control 4 Test 4 Eq. CPP L 6.0 6.0 6.0 6.31 6.0 6.2 6.0 6.0 TP by g/dL Eq CPP 0.57 0.52 0.53 0.54 0.60 0.65 0.60 0.62 Biuret IgG 4.51 4.02 4.04 4.03 4.67 4.82 4.52 4.54 Albumin 0.42 0.60 0.29 0.44 0.49 0.66 0.45 0.60 IgA 0.775 0.708 0.731 0.775 0.831 0.910 0.742 0.815 IgM g/L Eq CPP 0.223 0.189 0.148 0.112 0.150 0.254 0.145 0.175 C3 0.120 0.104 0.088 0.077 0.089 0.112 0.104 0.129 AAT 0.05 0.07 0.02 0.04 0.04 0.11 0.02 0.05 AMG 0.71 0.64 0.61 0.58 0.60 0.68 0.65 0.66 Fibrinogen ug/mL@5% TP 26 41 67.5 61.6 71.5 71.6 47.4 69.8

TABLE-US-00006 TABLE Precipitate G Protein Composition Control 1 Test 1 Control 2 Test 2 Control 3 Test 3 Control 4 Test 4 Eq. CPP L 6.0 6.0 6.0 6.31 6.0 6.2 6.0 6.0 TP by Biuret g/dL 5.97 6.13 6.12 6.17 6.81 6.34 5.9 6.0 IgG ug/mL@ 5% TP 4420 4200 3983 3993 4017 3893 4153 3795 Albumin 13.5 13.1 10.4 11.5 14.5 14.7 11.7 15.7 IgA 663 642 628 667 607 638 594 617 IgM 134 136 127 122 115 112 128 144 C3 74 75 85 73 75.6 69.8 83 97 AAT 6.08 5.98 5.13 6.79 8.40 9.93 4.62 10.64 AMG 398 390 345 345 282 321 311 336 Fibrinogen ug/mL@ 5% TP 29.2 28.3 61.8 60.3 77.0 76.2 47.2 75.4 FXI E/ml @5% TP 0.017 0.024 0.025 0.024 0.022 0.024 0.02 0.1 FXII mU/ml @5% 4.0 4.3 11.2 12.7 6.9 11.7 4.9 5.9 TP PL-1 nmol/mL min 13.1 8.2 15.1 19.2 18.2 14.8 12.6 15.2 @5% TP PKA IE/mL @5% 11.1 7.2 37 49 21 8.5 15.0 26.5 TP KKA nmol/mL min 119.0 75.9 118 133 106 80 96.3 122.3 @ 5% TP

[0103] Proteins typically found in the Fraction IV and V downstream from Cohn pool concentration are found in these fractions in yields and purity comparable to those found in Fraction IV and V in a process starting with non-concentrated Cohn Pool. The conditions of exemplary separation and purification processes for those plasma derived product intermediates produced from Cohn Pool concentrate are set forth in

Table—

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Table.

TABLE-US-00008 text missing or illegible when filed text missing or illegible when filed indicates data missing or illegible when filed

[0104] The supernatant of Fraction II+III was further submitted to fractionation procedure. The supernatant of Fraction II+III was contacted with 25% Ethanol to obtain Fraction IV-1 precipitate.

TABLE-US-00009 TABLE Protein step yields - Fraction IV-1 Filtrate Control 1 Test 1 Control 2 Test 2 Control 3 Test 3 Control 4 Test 4 Eq. CPP L 6.0 6.0 6.0 6.31 6.0 6.2 6.0 6.0 TP by g/dL Eq 2.80 2.39 2.94 2.73 2.70 2.62 2.70 2.70 Biuret CPP Albumin 27.1 23.6 26.1 23.9 25.2 24.6 27.3 25.8 TRF 0.88 0.77 1.08 1.05 0.64 0.64 0.96 0.78 AAT 0.039 0.029 0.015 0.017 0.016 0.019 0.060 0.036 HPT 0.59 0.50 0.58 0.57 0.56 0.53 0.66 0.59 AAG 0.59 0.50 0.58 0.52 0.54 0.53 0.60 0.56 PKA 7.9 20.1 222 330 57 8 59.0 29.9 KKA 27.8 15.3 441 604 201 79 185.1 135.6

TABLE-US-00010 TABLE Protein Step yields - Fraction IV-4 Filtrate Control 1 Test 1 Control 2 Test 2 Control 3 Test 3 Control 4 Test 4 Eq. CPP L 6.0 6.0 6.0 6.31 6.0 6.2 6.0 6.0 TP by Biuret g/dL Eq CPP 2.09 1.60 2.08 1.70 2.09 1.97 2.09 2.03 Albumin 22.3 17.9 21.1 17.1 20.6 19.7 22.0 20.7 TRF 0.03 0.02 0.03 0.016 AAT 0.01 0.01 0.01 0.008 HPT 0.05 0.02 0.23 0.30 0.08 0.06 0.10 0.10 AAG 0.56 0.45 0.50 0.40 0.50 0.49 0.53 0.50 CER 0.007 0.005 0.006 0.004

TABLE-US-00011 TABLE Fraction V Protein Composition Control Test Control Test Control Test Control Test 1 1 2 2 3 3 4 4 Eq. CPP L 6.0 6.0 6.0 6.31 6.0 6.2 6.0 6.0 TP by Biuret g/dL 6.04 6.19 6.05 6.19 5.97 6.01 5.7 6.1 Albumin ug/mL@ 5% TP 5548 5601 4509 4434 5212 5263 5306 5319 HPT 27.5 23.4 32.5 29.4 35.0 28.4 5306 5319 TRF 1.5 1.4 35 33 AAT 0.7 0.7 0.74 0.73 AAG 10.2 11.7 10.1 11.6 8.8 11.5 10.2 11.2 CER 0.38 0.37 PKA IE/mL @5% TP 3.3 4.0 8.3 18.5 23 7.9 14.3 16.8 KKA nmol/mL min @ 8.3 8.9 68 107 49 21 24.4 23.9 5% TP

Example 2

[0105] Cohn Pool Concentration (CPC) aims to increase capacity and reduce operational costs with minimal disruption to commercial supply. The process of present invention makes feasible of concentrating Cohn Pool plasma through ultrafiltration. Ultrafiltration permits the selective separation, concentration, and purification of protein components. Pilot scale runs have been performed in present invention and the desired Cohn Pool plasma volume and protein concentration were obtained without significant loss of various plasma derived proteins from different fractionations of Cohn Pool plasma.

[0106] Prior to Cohn fractionation process, about 100 liters cryo-poor plasma were concentrated in test runs (Run 1, Run 2, Run 3) using an ultrafiltration membrane while no ultrafiltration was performed in the control run. For the plasma concentration step, hollow-fiber filters with a nominal molecular weight cut off (NMWCO) of 100 kDa or less was employed. As a result, the starting protein concentrations of the Cohn Pool plasma in the test runs were about 15% higher than that in control run. For each protein of interest, technical specification of the separate and purified protein, for example, recovery rate, quality, are tabulated and analyzed.

1. Immunoglobulin Recovery Rate and Quality Were Not Affected by Cohn Pool concentration

[0107] Immunoglobulin content in all runs was calculated following the concentration step. The resulting IgG content in the Cohn Pool plasma prepared for Ethanol treatment is concentrated without significant loss. The Upstream IgG Efficiency in the test runs are comparable to the control run as shown in Table 1.

[0108] The recovery rate of purified IgG from Precipitate G (PptG) was also compared between test runs and control. Fraction II+III precipitate resulting from Cohn fractionation process were suspended in a suspension buffer, thereby forming an IgG suspension. The IgG suspension were filtered, and the filtrate were treated with detergent. After adjusting the pH of detergent-treated filtrate to about 7.0 and adding ethanol to a final concentration of from about 20% to about 30%, Precipitate G was formed. Table 2 shows that the IgG purification efficiency was not affected by the process of CPC. Comparable IgG yields from PptG between test runs and the non-concentrated control run have been observed in the pilot scale study. IgG quality was evaluated at drug product level for all runs. All results met all limits as shown in Table 3.

TABLE-US-00012 TABLE 1 Immunoglobulin balance in upstream processing Delta % (Run 1 vs.

Run 1 Run 1 Parameter/Step Unit Run 1 Control.sup.1 Control) Run 2 Run 3 Starting Material N/A Cohn Cohn Cohn Plasma Plasma Plasma after Cryo after after separation adsorption: adsorption: FEIBA.sup.2, Factor IX, AT III.sup.3 Factor VII Cohn Plasma (Starting [L] 100.5 100.5 112.7 120.3 Volume) Cohn Plasma for EtOH [L] 86.4 100 95.7 102.0 Frac Total Protein (pool [g/L] 58.9 (after 50.5 53.7 (63.9 53.7 (64.8 before conc.) conc.) after after conc.) conc.) IgG (pool before conc.) [g/L] 5.71 (after 4.92 5.07 (6.14 5.22 (6.10 conc.) after after conc.) conc.) Total IgG [g] 493 494 588 622 Extract Filtrate [L] 98.6 120.2 107.8 114.2 IgG [g/L] 4.67 3.71 4.96 4.85 Total IgG [g] 460.3 445.9 +3.2% 534.6 553.8 IgG g/L plasma before [g/L] 4.55 4.40 +3.2% 4.74 4.53 conc Upstream IgG [%] 93.1 90.2 +3.2% 93.5 88.2 Efficiency .sup.1Same starting material was used for ‘Run 1 Control’ and ‘Run 1’, i.e., Cohn Plasma after Cryo separation was split and processed without Cohn Pool Concentration (‘Run 1 Control’) and with Cohn Pool Concentration (‘Run 1’) in parallel. .sup.2Factor eight inhibitor bypass activity .sup.3Antithrombin III TABLE-US-00013 TABLE 2 Immunoglobulin Balance in Test Runs and Control Run after Purification Delta % Run 1 (Run 1 vs. Run 1 Control Run 1 Unit Ppt G Ppt G Control) Run 2 Run 3 CPP L 100.5 100.5 0.0 112.7 120.3 PptG weighed in kg 1.744 1.778 2.343 2.199 fractionation Retrieved for kg 0.996 1.002 1.000 1.004 suspension TP in suspension g 310.0 288.6 7.4 283.3 307.7 IgG in suspension g 264.3 244.9 7.9 222.3 251.0 IgG per kg of paste g IgG/kg 265.3 244.4 8.5 222.2 250.0 PptG IgG in paste/CPP Eq g IgG/L 4.60 4.32 6.4 4.62 4.57 CPP TP loaded on filter g 260 260 260 260 TP loaded on filter g 221 220 212 221 (minus losses) IgG loaded on filter g 226.8 225.8 0.4 210.9 216.9 IgG loaded on filter g 193.9 192.1 0.9 170.7 184.0 (minus losses) PptG Eq loaded on kg 0.836 0.903 −7.4 0.918 0.848 filter PptG Eq loaded on kg 0.710 0.763 −7.0 0.749 0.720 filter (minus losses) CPP Eq L 48.2 51.0 −5.6 44.2 46.4 CPP Eq L 40.9 43.2 −5.2 36.0 39.4 (minus losses) IgG After CM g 176.8 182.5 173.5 172.6 IgG CM/Suspension % 78.0 80.8 −3.5 82.3 79.6 Yield IgG CM/Suspension % 86.0 89.5 −3.9 94.1 87.9 Yield (incl losses) Final solution weight kg 1.6551 1.6756 1.6196 1.6699 Recovered IgG g 160.4 162.4 −1.2 156.6 161.6 IG Efficiency % 70.7 71.9 −1.7 74.3 74.5 (Purification Efficiency) IG Efficiency % 82.7 84.5 −2.1 91.7 87.9 (Purification Efficiency incl. losses) Recovered IgG/CPP Eq g IgG/L 3.33 3.18 4.7 3.55 3.48 CPP Recovered IgG/CPP Eq g IgG/L 3.92 3.76 4.2 4.35 4.10 (incl losses) CPP TABLE-US-00014 TABLE 3 Immunoglobulin Drug Product Test Results Acceptance Run 1 Test Criteria Units Run 1 Control Run 2 Run 3 Diphtheria ≥1.2 U of U/mL 10.4 9.8 9.8 12.2 Antibody US standard Antitoxin/mL Poliomyelitis ≥0.2 times the 1.1 0.9 1.0 1.2 Antibody antibody level of CBER Reference Polio Immune Globulin Lot 176 IgA ≤0.14 mg/mL 0.06 0.06 0.06 0.07 IgM ≤10 mg/dL <4 <4 <4 <4 Anti-A NMT 1/64 1/ 32 16 16 16 @25 g/L of IgG Anti-B NMT 1/64 1/ 8 8 16 16 @25 g/L of IgG Anti-D NMT NIBSC — Satisfactory Satisfactory Satisfactory Satisfactory antibodies Ref. 02/228 (or equivalent) Anti- NMT 50 % 33 38 34 34 complement corresponding Activity to 1 CH50 U/mg protein Total Protein 9.0-11.0 g/dL 10.0 9.9 10.0 10.0 Appearance The liquid — Satisfactory Satisfactory Satisfactory Satisfactory preparation is clear or slighlty opalescent and colorless or pale yellow Glycine 0.20-0.30/ M 0.23 0.23 0.23 0.24 0.21-0.26 HAV Antibody ≥3.5 IU/mL 5.8 8.2 5.9 6.0 HBsAg ≥0.20 IU/mL 8.17 7.61 7.09 6.92 Antibody Molecular Size Monomer + % 100 100 100 100 Distribution Dimer: ≥95 Polymer: ≤2 % 0.1 0.1 0.1 0.1 Fragment: ≤3 % 0.3 0.2 0.2 0.2 Purity ≥98 % 100 100 100 100 Osmolality 240-300 mOsmol/ 268 269 263 264 kg Parvo B19 ≥50 IU/mL 414 459 441 367 pH 4.6-5.1 — 4.7 4.7 4.7 4.7 Diluted @ 1% with 0.9% NaCl Prekallikrein ≤10 — <5 <5 <5 <5 Activator Activity Tween 80 ≤100 ppm <26 <26 <26 <26 TNBP ≤1.0 ppm <0.2 <0.2 <0.2 <0.2 Triton X-100 ≤1.0 ppm <0.1 <0.1 <0.1 <0.1 Density N/A g/cm.sup.3 1.03 1.03 1.03 1.03 THP-1/IL-1ra NLT 75% % 115 128 137 126 Release Ig G 1 N/A mg/mL 64.0 65.6 Subclass 52.6-68.4 % 62.4 62.4 58.6 59.2 Distribution Ig G 2 N/A mg/mL 30.9 32.0 Subclass 23.5-40.5 % 30.2 30.5 33.3 32.9 Distribution Ig G 3 N/A mg/mL 5.0 4.8 Subclass 2.9-7.8 % 4.8 4.6 5.1 5.1 Distribution Ig G 4 N/A mg/mL 2.7 2.6 Subclass 1.2-3.2 % 2.6 2.5 3.0 2.8 Distribution IgE ≤149.6 IU/mL 53.9 52.7 42.0 48.6 Complement- ≥60 (cfr. EP) % 107 106 91 84 mediated Lysis of Red Blood Cells Haemophilus ≥1:1600 — 1:12800 1:12800 1:12800

1:12800 Influenzae Ab Albumin <0.22 g/L <0.22 <0.22 <0.222 <0.222 PL-1 <10 nmol/mL × <10 <10 <10 <10 Amidolytic min activity Fibrinogen ≤1.5 μg/mL <0.2 <0.2 <0.2 <0.2 Plasminogen ≤0.3 μg/mL <0.12 <0.12 <0.12 <0.12 Alcohol <20 μg/mL <20 <20 <20 <20 (Ethanol) Aluminum ≤34 μg/L <25 <25 <25 <25 FXIa Control mU/mL 0.47 <0.3125 <0.3125 <0.3125 limit: ≤1.6 mU/mL FXIa Action limit: ≤6.0 mU/mL FXIa FXIa @5% Control mU/mL <0.3125 <0.3125 <0.3125 <0.3125 limit: ≤1.6 mU/mL FXIa Action limit: ≤6.0 mU/mL FXIa FXI Protein N/A U/mL 0.11 0.04 0.02 0.01 TGA(Thrombin <132% NP % NP 120 109 110 105 Generation equivalent to Assay) 1.7 mU/mL FXIa

2. Process Parameters for Immunoglobulin Separation and Purification From Cohn Pool

[0109] The conditions for an exemplary plasma derived immunoglobulin proteins separation process, including Cohn Pool concentration, fraction I precipitation and separation, fraction II+III precipitation and separation, fraction II+III extract precipitation and separation, precipitate G (PptG) precipitation and separation, are set forth in Table 4-8.

[0110] In the exemplary CPC process, target amounts of filter aid and filter area were calculated based on the volume of concentrated Cohn Pool instead on the volume of cryo-poor plasma (CPP). Cohn Pool was concentrated of at least about 14% (w/v) by ultrafiltration, forming a first Cohn Pool concentrate. The technical protocol and parameters associated with the process of ultrafiltration in all runs were indicated in Table 4.

TABLE-US-00015 TABLE 4 Parameters for Cohn Pool Concentration Step Description Unit Run 1 Run 2 Run 3 Cohn Pool mass [kg] 102.6 115.1 122.8 before concentration Cohn Pool volume [L] 100.5 112.7 120.3 before concentration Type of filter for N/A hollow- fiber hollow- fiber hollow- fiber concentration 100 kDa MWCO 100 kDa MWCO 100 kDa MWCO Filter surface area [m.sup.2] 0.7 1.8 1.8 Number of filters used N/A 1 1 1 Filtration time [hh:mm] 05:31 02:58 02:44 Maximum sustained TMP [bar] 0.52 0.46 0.42 Average retentate flowrate [L/min] 1.08 1.69 1.73 Average permeate flowrate [g/min] 50.3 110.7 119.9 Permeate weight [kg] 16.66 19.70 19.66 Initial concentration [%] 16.24% 15.87% 16.01% factor (by mass) Post-wash using 0.9% NaCl [kg] 0.415 0.579 0.576 Cohn Pool mass after [kg] 88.2 105.6 104.4 concentration & post-wash Cohn Pool volume after [L] 86.4 95.7 102.0 concentration & post-wash Final concentration [%] 14.04% 14.91% 14.98% factor (by mass) Filter load [L CPP/m.sup.2] 143.6 62.6 66.8 Turbidity - before conc./ [NTU] 223/249 262/288 270/309 after conc.

[0111] The first Cohn Pool concentrate was further submitted to fractionation procedure. The first Cohn Pool concentrate was contacted with 8% ethanol at a pH of from about 7.0 to 7.5 to obtain a Fraction I precipitate and a Fraction I supernatant from the fractionated Cohn Pool. The technical protocol and parameters associated with fraction I precipitation and separation in all runs were indicated in Table 5.

TABLE-US-00016 TABLE 5 Parameters of Fraction I Precipitation and Separation with 8% Ethanol Run 1 Parameter Unit Run 1 Control Run 2 Run 3 pH Before Alcohol [pH] 7.21 7.19 7.01 7.0 Temperature Before [° C.] 0.2 0.0 1.1 0.9 Alcohol Alcohol Add Time [hh:mm] 01:00 01:01 01:00 01:00 pH After Alcohol [pH] 7.38 7.38 7.17 7.17 Temperature After [° C.] −0.8 −0.5 −0.4 −0.3 Alcohol Final pH [pH] 7.51 .fwdarw. 7.18 7.47 .fwdarw. 7.21 7.41 7.32 Total Aging Time [hh:mm] 13:51 13:51 15:53 17:25 Suspension Weight [kg] 94.7 110.2 113.3 112.1 Separation Method N/A CEPA Z61H CEPA Z61H CEPA Z61H CEPA Z61H Supernatant Density [kg/L] 1.008 1.006 1.010 1.010 Total Supernatant [L] 92.7 108.1 103.1 109.2 Volume Paste Weight [kg] 0.709 0.704 1.081 1.047 Duration of [hh:mm] 02:03 02:21 02:36 01:20 Centrifugation Average Flowrate [L/h] 45.2 46.0 39.7 81.9

[0112] The Fraction I supernatant was further contacted with about 25% ethanol at a pH of from about 6.7 to about 7.3 to form a Fraction II+III precipitate. The technical parameters associated with isolating proteins from fraction II+III of fractionated Cohn Pool in all runs were indicated in Table 6.

TABLE-US-00017 TABLE 6 Parameters of Fraction II + III Precipitation and Separation with 25%

Ethanol Run 1 Parameter Unit Run 1 Control Run 2 Run 3 pH Before Alcohol [pH] 6.71 6.67 6.68 6.73 Temperature Before [° C.] −0.9 −0.9 −1.0 −1.9 Alcohol Alcohol Add Time [hh:mm] 03:00 03:00 03:00 03:00 pH After Alcohol [pH] 7.18 .fwdarw. 6.92 7.05 .fwdarw. 6.89 7.08 .fwdarw. 6.89 7.20 .fwdarw. 6.89 Temperature After [° C.] −6.6 −6.5 −6.5 −6.6 Alcohol Total Aging Time [hh:mm] 16:31 16:25 17:11 18:14 Final pH [pH] 7.09 6.96 7.34 .fwdarw. 6.90 7.33 .fwdarw. 6.91 Suspension Weight [kg] 111.6 129.9 124.4 131.7 Separation Method N/A Eaton 400 Eaton 400 Eaton 400 Eaton 400 filter press filter press filter press filter press Filter Type N/A Becopad Becopad Becopad Becopad P270 P270 P270 P270 Effective Surface Area [m.sup.2] 1.21 1.32 1.32 1.43 Filter Load [L CP/m.sup.2] 83.1 76.1 85.4 84.1 Frame Thickness (single) [mm] 25 25 25 25 Frame Volume (single) [L] 2.20 2.20 2.20 2.20 Dead volume [L] 13.1 14.2 14.2 15.3 Filteraid Type N/A Fibracel Fibracel Fibracel Fibracel Filteraid Target [g/L] 22 g/L CPC 22 g/L CPP 22 g/L CPC 22 g/L CPC Filteraid Amount [kg] 1.901 2.200 2.105 2.243 Pre-Rinse Vol (water) [L] 42.4 46.2 46.2 50.1 Pre-Wash Vol (wash [L] 59.1 63.2 68.3 62.1 solution) Post-Wash Vol (wash [L] 13.1 14.2 14.2 15.3 solution) Temp (end of pre-wash) [° C.] −2.6 −2.8 −4.0 −3.5 Mean Filtration Temp [° C.] −4.9 −5.0 −5.2 −5.1 Max Sustained Pressure [bar] 1.0 0.9 1.0 1.1 (filtration) Final Flux [L/hr/m.sup.2] 42.70 47.87 24.91 25.0 Supernatant Density [kg/L] 0.979 0.979 0.981 0.981 Total Supernatant Volume [L] 129.2 150.4 142.4 149.4 Paste Weight [kg] 7.822 8.58 10.07 10.512 Duration of Filtration [hh:mm] 01:53 01:48 03:11 03:17 Duration of Filtration + [hh:mm] 02:30 02:23 04:20 04:19 Post-Wash

[0113] The Fraction II+III precipitate was further suspended in a suspension buffer, thereby forming an IgG suspension. After mixing finely divided silicon dioxide (SiO.sub.2) with the IgG suspension for at least about 30 minutes, IgG suspension was filtered and resulted a filtrate and a filter cake. The technical parameters associated with isolating proteins from fraction II+III extract in all runs were indicated in Table 7.

TABLE-US-00018 TABLE 7 Parameters of Fraction II + III Extract Precipitation and Separation Run 1 Parameter Unit Run 1 Control Run 2 Run 3 Theoretical Wet Paste [kg] 4.29 4.29 4.78 5.14 Yield Suspension pH (Initial) [pH] 5.0 5.1 4.99 4.96 Temperature of Suspension [° C.] 3.9 4.0 4.1 3.9 Acetic Acid in Extraction [g/L] 0.42 0.32 0.42 0.42 Buffer Extraction Buffer Weight [kg] 64.5 83.8 71.7 77.3 Buffer-to-Theoretical N/A 15.0 19.5 15.0 15.0 Paste Ratio Extraction Time [hh:mm] 03:00 03:00 03:00 03:00 Final pH [pH] 5.01 5.09 5.01 5.02 Suspension Weight [kg] 72.3 92.4 81.8 87.9 Separation Method N/A Eaton 400 Eaton 400 Eaton 400 Eaton 400 filter press filter press filter press filter press Filter Type N/A Cuno 50SA Cuno 50SA Cuno 50SA Cuno 50SA Effective Surface Area [m.sup.2] 0.88 0.99 0.99 0.99 Filter Load [L CP/m.sup.2] 98.2 101.0 113.9 121.5 Frame Thickness (single) [mm] 30 30 30 30 Frame Volume (single) [L] 2.75 2.75 2.75 2.75 Dead volume [L] 12.0 13.4 13.4 13.4 Aerosil Target [g/L CPP] 2.3 2.3 2.3 2.3 Aerosil Amount [kg] 0.2303 0.2302 0.2549 0.2750 Aerosil Contact Time [hh:mm] 03:27 02:23 02:11 02:01 Filteraid Type N/A Celpure C300 Celpure C300 Celpure C300 Celpure C300 Filteraid Target [g/L] 12 g/L CPC 12 g/L CPP 12 g/L CPC 12 g/L CPC Filteraid Amount [kg] 1.0368 1.2003 1.1495 1.2249 Pre-Wash Vol (wash [L] 81.0 88.2 83.2 83.2 solution) Post-Wash Vol (wash [L] 24.0 26.8 26.8 26.8 solution) Temp (end of pre-wash) [° C.] 5.0 5.4 4.9 4.7 Mean Filtration Temp [° C.] 4.6 4.5 4.3 4.5 Max Pressure (filtration) [bar] 1.0 1.0 1.1 1.0 Final Flux [L/hr/m.sup.2] 52.7 115.4 101.6 135.5 Supernatant Density [kg/L] 0.999 0.999 0.999 0.999 Total Supernatant Volume [L] 98.5 120.1 107.7 114.1 Duration of Filtration [hh:mm] 01:20 00:45 00:32 00:30 Duration of Filtration + [hh:mm] 02:07 01:03 01:04 00:51 Post-Wash

[0114] The filtrate resulting from Fraction II+III precipitate was contacted with a detergent. forming a treated filtrate. After adjusting the pH of the treated filtrate to about 7.0, and adding ethanol to a final concentration of from about 20% to about 30%. a Precipitate G precipitate was formed. The technical parameters associated with Ppt G precipitation and separation from Fraction II+III precipitate in all runs were indicated in Table 8.

TABLE-US-00019 TABLE 8 Parameters of Ppt G Precipitation and Separation Run 1 Parameter

Unit Run 1 Control Run 2 Run 3 pH Before Adjustment [pH] 6.55 6.36 6.38 6.40 pH Before Alcohol [pH] 6.95 7.01 7.00 7.01 Temperature Before [° C.] 1.5 2.0 1.1 1.8 Alcohol Alcohol Add Time [hh:mm] 04:00 04:00 04:01 04:01 Temperature After [° C.] −9 −9 −9 −9 Alcohol Final pH [pH] 7.20 7.23 7.11 7.07 Total Aging Time [hh:mm] 10:11 10:11 12:05 12:14 Suspension Weight [kg] 128.0 156.1 140.0 148.3 Separation Method N/A CEPA Z61H CEPA Z61H CEPA Z61H CEPA Z61H Supernatant Density [kg/L] 0.971 0.971 0.971 0.971 Total Supernatant [L] 126.5 157.4 141.3 149.9 Volume Paste Weight [kg] 1.7436 1.7775 2.343 2.199 Duration of [hh:mm] 02:44 03:26 01:46 01:53 Centrifugation Average Flowrate [L/h] 46.3 45.8 80.0 79.6 (approx)

3. Plasma Derived Products Isolated From Fraction IV and V of Fractioned Cohn Pool Were Not Affected by Cohn Pool Concertation

[0115] Proteins typically found in the Fraction IV and V downstream from Cohn pool concentration are found in these fractions in yields and purity comparable to those found in Fraction IV and V in a process starting with non-concentrated Cohn Pool. The conditions of exemplary separation and purification processes for those plasma derived products from Cohn Pool concentrate, including Albumin, Alpha-1 proteinase inhibitor, are set forth in Table 9-14.

[0116] The supernatant of Fraction II+III was further submitted to fractionation procedure. The supernatant of Fraction II+III was contacted with 25% ethanol to obtain Fraction IV-1 precipitate. The technical protocol and parameters for isolating protein in Fraction IV-1 were indicated in Table 9.

TABLE-US-00020 TABLE 9 Parameters of Fraction IV-1 Precipitation and Separation with 25% Ethanol Run 1 Parameter Unit Run 1 Control Run 2 Run 3 pH Before Aging [pH] 5.56 5.51 5.47 5.49 Suspension Temp [° C.] −6.5 −6.5 −5.5 −5.5 Total Aging Time [hh:mm] 13:40 13:40 13:09 14:07 Final pH [pH] 5.51 5.47 5.41 5.48 Suspension Weight [kg] 127.3 148.1 140.3 147.1 Separation Method N/A Eaton 400 Eaton 400 Eaton 400 Eaton 400 filter press filter press filter press filter press Filter Type N/A Becopad Becopad Becopad Becopad P270 P270 P270 P270 Effective Surface Area [m.sup.2] 0.55 0.66 0.66 0.77 Filter Load [L CP/m.sup.2] 182.7 152.3 170.8 156.2 Frame Thickness (single) [mm] 25 25 25 25 Frame Volume (single) [L] 2.20 2.20 2.20 2.20 Dead volume [L] 6.5 7.6 7.6 8.7 Filteraid Type N/A Celpure C300 Celpure C300 Celpure C300 Celpure C300 Filteraid Target [g/L] 13 g/L CPC 13 g/L CPP 13 g/L CPC 13 g/L CPC Filteraid Amount [kg] 1.1233 1.3002 1.243 1.3263 Pre-Rinse Vol (water) [L] 13.8 16.5 16.5 19.3 Pre-Wash Vol (wash [L] 49.5 59.4 66.7 70.0 solution) Post-Wash Vol (wash [L] 10.5 12.5 12.5 14.6 solution) Temp (end of pre-wash) [° C.] −4.0 −4.0 −3.9 −4.4 Mean Filtration Temp [° C.] −5.0 −5.2 −4.7 −4.7 Max Sustained Pressure [bar] 1.5 0.9 1.0 0.9 (filtration) Final Flux [L/hr/m.sup.2] 57.1 64.5 71.6 70.0 Supernatant Density [kg/L] 0.978 0.977 0.979 0.979 Total Supernatant Volume [L] 147.8 165.0 156.7 165.5 Paste Weight [kg] 4.918 5.756 5.525 5.948 Duration of Filtration [hh:mm] 03:55 03:18 02:55 02:33 Duration of Filtration + [hh:mm] 04:46 03:56 03:22 03:07 Post-Wash

[0117] The supernatant of Fraction IV-1 was subsequently contacted with 40% ethanol to obtain Fraction IV-4 precipitate. The technical parameters for isolating proteins in Fraction IV-4 were indicated in Table 10.

TABLE-US-00021 TABLE 10 Parameters of Fraction IV-4 Precipitation and Separation with 40% Ethanol Run 1 Parameter Unit Run 1 Control Run 2 Run 3 pH Before ISA [pH] 5.54 5.50 5.50 5.49 pH Before Alcohol [pH] 5.76 5.79 5.70 5.88 Temperature Before [° C.] −6.6 −6.3 −6.0 −6.0 Alcohol Alcohol Add Time [hh:mm] 05:00 05:05 05:00 05:00 pH After Alcohol [pH] 5.82 .fwdarw. 5.96 5.82 .fwdarw. 5.96 5.80 .fwdarw. 5.93 5.83 .fwdarw. 5.94 Temperature After [° C.] −5.8 −6.2 −6.1 −6.0 Alcohol Total Aging Time [hh:mm] 07:46 07:57 10:23 10:29 Final pH [pH] 5.96 5.94 5.95 5.97 Suspension Weight [kg] 179.5 201.1 192.1 202.41 Separation Method N/A Eaton 400 Eaton 400 Eaton 400 Eaton 400 filter press filter press filter press filter press Filter Type N/A Cuno 70CP Cuno 70CP Cuno 70CP Cuno 70CP Effective Surface Area [m.sup.2] 0.66 0.77 0.77 0.88 Filter Load [L CP/m.sup.2] 152.3 130.5 146.4 136.7 Frame Thickness (single) [mm] 25 25 25 25 Frame Volume (single) [L] 2.20 2.20 2.20 2.20 Dead volume [L] 7.6 8.7 8.7 9.8 Filteraid Type N/A

Celpure C300 Celpure C300 Celpure C300 Celpure C300 Filteraid Target [g/L] 15.5 g/L CPC 17.2 g/L CPP 15.5 g/L CPC 15.5 g/L CPC Filteraid Amount [kg] 1.339 1.720 1.482 1.582 Pre-Rinse Vol (water) [L] 14.9 17.3 17.3 19.8 Pre-Wash Vol (wash [L] 47.8 59.7 57.9 59.4 solution) Post-Wash Vol (wash [L] 15.2 17.4 17.4 19.6 solution) Temp (end of pre-wash) [° C.] −4.4 −4.4 −4.6 −5.1 Mean Filtration Temp [° C.] −5.4 −5.6 −5.3 −5.4 Max Sustained Pressure [bar] 0.9 0.9 1.0 1.0 (filtration) Final Flux [L/hr/m.sup.2] 87.0 93.8 76.9 104.3 Supernatant Density [kg/L] 0.953 0.954 Not measured Not measured Total Supernatant Volume [L] 196.0 223.9 ~217.2 ~230.0 Paste Weight [kg] 4.059 4.376 4.803 4.861 Duration of Filtration [hh:mm] 02:54 02:40 03:10 02:55 Duration of Filtration + [hh:mm] 03:23 03:04 03:38 03:29 Post-Wash

[0118] The supernatant of Fraction IV-4 was subsequently contacted with 40% ethanol to obtain Fraction V precipitate. Fraction V of the invention includes primarily albumin. The technical protocol and parameters for isolating proteins in Fraction V of fractionated Cohn Pool in all runs were indicated in Table 11. Comparable total protein content in fraction IV-4 between Run 1 and the non-concentrated control run have been observed, as shown in Table 12.

TABLE-US-00022 TABLE 11 Parameters of Fraction V Precipitation and Separation with 40% Ethanol Run 1 Parameter Unit Run 1 Control Run 2 Run 3 pH Before Buffer Add [pH] 5.90 5.87 5.94 5.97 Temperature Before [° C.] −6.6 −6.4 −6.4 −6.1 Buffer Add Buffer Add Time [hh:mm] 06:00 06:00 06:00 06:00 Total Aging Time [hh:mm] 10:27 10:27 12:34 13:15 Final pH [pH] 4.80 4.78 4.78 4.79 Temperature After Aging [° C.] −6.8 −6.8 −6.8 −6.9 Suspension Weight [kg] 191.0 218.4 211.0 223.8 Separation Method N/A Eaton 400 Eaton 400 Eaton 400 Eaton 400 filter press filter press filter press filter press Filter Type N/A Becopad Becopad Ahlstrom Ahlstrom P270 P270 950 950 Effective Surface Area [m.sup.2] 0.66 0.77 0.99 1.10 Filter Load [L CP/m.sup.2] 152.3 130.5 113.9 109.3 Frame Thickness (single) [mm] 30 30 30 30 Frame Volume (single) [L] 2.75 2.75 2.75 2.75 Dead volume [L] 9.25 10.6 13.4 14.8 Filteraid Type N/A N/A N/A N/A N/A Pre-Rinse Vol (water) [L] 14.9 17.3 N/A N/A Pre-Wash Vol (wash [L] 50.6 59.1 66.9 83.3 solution) Post-Wash Vol (wash [L] N/A N/A N/A N/A solution) Temp (end of pre-wash) [° C.] −4.9 −5.1 −6.0 −6.0 Mean Filtration Temp [° C.] −5.4 −5.8 −5.9 −5.9 Max Sustained Pressure [bar] 2.4 2.4 2.0 2.2 (filtration) Final Flux [L/hr/m.sup.2] 47.4 53.1 109.6 39.4 Supernatant Density [kg/L] 0.950 0.951 0.951 0.952 Total Supernatant Volume [L] 143.4 187.2 219.3 232.0 Remaining Suspension [kg] 54.8 40.3 N/A N/A Weight .sup.(1) Paste Weight [kg] 6.911 7.761 11.852 13.189 Calculated Total Paste .sup.(1) [kg] 9.693 9.518 N/A N/A Duration of Filtration [hh:mm] 04:35 04:35 02:01 05:21

TABLE-US-00023 TABLE 12 Balance of Protein Isolated from Fraction IV-4 Run 1 Parameters Run 1 Control Delta Cohn Plasma (Starting Volume) L 100 100 Cohn Plasma for EtOH Frac L 86.4 100 Total Protein (pool before conc.) g/l 50.5 Fr. IV4 Supern. Filtrate L 196.1 223.9 Fr. IV4 Supern. Filtrate TP g/L 12.9 11.5 Fr. IV4 Supern. Filtrate Total Protein g 2529 2575 Fr. IV4 Supern. Filtrate g/L Alb./TP CP 25.3 25.7 −1.56% (before conc.)

[0119] The quality of proteins isolated from Fraction V were evaluated through multiple biochemical and immunological assays as shown in Table 13. Quality indicating parameters of isolated proteins in Fraction V of Run 1 are aligned with the results of the respective non-concentrated control.

TABLE-US-00024 TABLE 13 Extended Characterization Testing at Fraction V Test Parameter Unit Run 1 Run 1 Control Albumin (Neph) mg/mL 52.250 52.700 Haptoglobin μg/mL 263 330 MSD - HPLC (Aggregates) % area 0.48 0.72 MSD - HPLC (Dimers) % area 8.81 10.14 MSD - HPLC (Fragments) % area 0.26 0.31 MSD - HPLC (Monomers) % area 90.44 88.83 PKA IE/mL <4 <4 Purity % 97.2 97.6 Total Protein (Biuret) mg/mL 51.4 50.9 Transferrin μg/mL 5.82 5.00 al-Antitrypsin mg/mL <0.045 <0.045

[0120] Fraction IV-1 intermediate isolated from concentrated Cohn Pool plasma was further processed until A1PI drug product. All relevant processing parameters are summarized in Table 14.

TABLE-US-00025 TABLE 14 Parameters of A1PI isolation and purification from Fraction IV-1

Run 1/ Run 1 Run 1 Parameter Unit Run 1 Control Control % Run 2 Run 3 CPP L 100.5 100.5 0.0 112.7 120.3 FrIV-1 weighed in g 4197.2 5756.0 −27.1 5525.0 5948.0 fractionation FrIV-1 suspended g 4165.7 4950.9 −15.9 4966.0 4969.9 CPP Eq L 99.7 86.4 15.4 101.3 100.5 TP in suspension g 863.8 904.8 −4.5 764.3 710.0 AAT in suspension g 55.2 35.9 53.8 56.0 53.5 AAT in suspension/CPP Eq g AAT/L 0.553 0.415 33.3 0.553 0.532 CPP AAT per kg of paste g AAT/kg 13.25 7.25 82.7 11.28 10.76 FrIV-1 Recovered AAT g 34.7 25.7 35.0 38.7 38.0 after DEAE-1 Recovered AAT/CPP Eq g AAT/L 0.349 0.298 17.0 0.383 0.379 after DEAE-1 CPP DEAE-1 Step Yield % 74.92 79.61 −5.9 81.84 84.43 Global Yield % 63.09 71.85 −12.2 69.23 71.15 Recovered AAT g 34.5 20.3 70.0 30.5 31.1 after CM Recovered AAT/CPP Eq g AAT/L 0.347 0.236 47.3 0.302 0.310 after CM CPP CM Step Yield % 95.52 80.40 18.8 84.67 87.41 Global Yield % 63.33 57.37 10.4 55.08 58.84 Recovered AAT g 26.7 14.9 79.2 23.2 23.9 after DEAE-2 Recovered AAT/CPP Eq g AAT/L 0.269 0.173 55.3 0.230 0.239 after DEAE-2 CPP DEAE-2 Step Yield % 82.45 82.33 0.1 81.25 82.34 Global Yield % 50.08 43.92 14.0 42.93 46.41 Recovered AAT g 23.2 10.8 114.8 18.6 19.3 at Drug Substance Recovered AAT/CPP Eq g AAT/L 0.236 0.127 86.4 0.186 0.195 at Drug Substance CPP Drug Substance % 96.51 88.61 8.9 88.41 92.09 Step Yield Global Yield until % 45.4 33.6 35.3 35.8 39.0 Drug Substance

[0121] Human Albumin and A1PI isolated from concentrated Cohn Pool plasma meet all predefined limits. All results of additional testing performed at drug product and intermediate level are within predefined ranges as shown in Table 15-17.

[0122] All results met the predefined acceptance criteria at intermediate and drug product level. Based on the illustrated results it can be concluded that Cohn Pool concentration does not negatively impact product quality of either A1PI or Albumin.

TABLE-US-00026 TABLE 15 Albumin Drug Product Test Results for Albumin Run 1 and Run 1 Control Run 1 Run 1 Control Acceptance Before After Before After Test Criteria Units Pasteurization Pasteurization Pasteurization Pasteurization Physical Clear, slightly Pass Pass Pass Pass Appearance viscous liquid, almost colorless, yellow, amber, or green Total 4.70 to 5.30 g/dL 5 5 4.9 4.9 Protein g/100 mL Protein Albumin: 96% % 99% 99% 99% 99% Composition minimum, (PCA) Globulins: 4% 1% 1% 1% 1% maximum N- 0.064 to 0.096 mmol/g 0.083 0.083 0.081 0.08 Acetyl mmol/g protein Tryptophan protein Octanoic 0.064 to 0.096 mmol/g 0.067 0.067 0.078 0.077 Acid mmol/g protein protein Sodium 130 to 160 mEq/L 152 152 149 151 mEq/L Potassium 2 mEq/L mEq/L <0.6 <0.6 <0.6 <0.6 maximum Turbidity Initial (heated NU initial 8 nu initial 8 nu initial 8 nu initial 9nu 10 to 11 hrs. at heated 9 nu heated 9 nu heated 9 nu heated 9 nu 60° C. ± 0.5° C.): 30 N.U. maximum Heated (50 hrs. at 57° C.): 30 N.U. maximum Heat Shall remain SAT SAT SAT SAT Stability visually unchanged after heating at 57° C. for 50 hours when compared to an unheated sample from the same lot Heme A403 nm = Absorbance 0.04 0.03 0.04 0.03 Content 0.25 maximum unit PKA IU/mL N/A <4 N/A <4 Ph 6.85 to 7.05 7 6.9 6.8 6.8 (6.4 to 7.4)

TABLE-US-00027 TABLE 16 Albumin Drug Product Test Results for Albumin Run 2 and Run 3 Run 2 Run 3 Before After Before After Test Acceptance Criteria Units Pasteurization Pasteurization Pasteurization Pasteurization Physical Clear, slightly viscous liquid, Pass Pass Pass Pass Appearance almost colorless, yellow, amber, or green Total Protein 4.70 to 5.30 g/100 mL g/dL 5 5 4.9 4.9 Protein Albumin: 96% minimum, % 99% 99% 99% 99% Composition Globulins: 4% maximum 1% 1% 0% 1% (PCA) N-Acetyl 0.064 to 0.096 mmol/g protein mmol/g 0.085 0.085 0.081 0.08 Tryptophan protein Octanoic 0.064 to 0.096 mmol/g protein mmol/g 0.073 0.073 0.077 0.076 Acid protein Sodium 130 to 160 mEq/L mEq/L 154 149 147 145 Potassium 2 mEq/L maximum mEq/L <0.6 <0.6 <0.6 <0.6 Turbidity Initial (heated 10 to 11 hrs. at NU initial 9 nu initial 10 nu initial 9 nu initial 10 nu 60° C. ± 0.5° C.): 30 N.U. heated 10 nu heated 11 nu heated 10 nu heated 11 nu maximum Heated (50 hrs. at 57° C.): 30 N.U. maximum Heat Stability Shall remain visually SAT SAT SAT SAT unchanged after heating at 57° C. for 50 hours when compared to an unheated sample from the same lot Heme A403 nm = 0.25 maximum Absorbance 0.05 0.05 0.07 0.06 Content unit PKA IU/mL <4 <4 <4 <4 Ph 6.85 to 7.05 (6.4 to 7.4) 6.9 6.9 7.0 7.0

TABLE-US-00028 TABLE 17 A1PI Drug Product Test Results Acceptance Run 1 Test Criteria Units Run 1 Control Run 2 Run 3 AAT Concentration 16-24 mg/mL mg/mL 21 22 21 21 AAT Activity 0.7-1.1 mg mg 1.0 1.0 1.0 1.0 AAT/mg protein AAT/mg protein Total Protein 16-24 mg/mL mg/mL 20 23 21 21 Gel Electrophoresis Co migrates with Pattern conform to conform to conform to conform to API Standard reference reference reference reference profile profile profile profile Molecular Size Monomers NLT % 100 100 100 100 Distribution 95% <1.0 <0.3 <0.3 <0.3 Non-monomeric <0.5 <0.5 <0.5 <0.5 forms NMT 5% Fragments NMT 1.6% S/D Detergents Residual Tween ppm ≤20 ≤20 ≤20 ≤20 NMT 20 ppm ≤5 ≤5 ≤5 ≤5 Residual TNBP NMT 5 ppm Residual PEG (UV NMT 20 ppm ppm ≤20 ≤20 ≤20 ≤20 spectroscopy) Appearance/ Solution is clear N/A satisfactory satisfactory satisfactory satisfactory Visual Inspection and colorless to yellow-green May contain a few particles pH 6.8-7.2 — 7.0 7.0 7.0 7.0 Sodium 130-170 mEq/L 168 161 160 154 (Sodium) Chloride 6-8 mg/mL 7 7 7 7 Phosphate 18-22 mM 21 21 21 21 Isoelectric focus Similar pattern to Pattern conform to conform to conform to conform to Gel API Standard reference reference reference reference profile profile profile profile Total Protein 16-24 mg/mL 21 22 20 21 (Bradford)

Example 3

[0123] A commercial scale run has been performed in present invention until Precipitate G (PptG), Fraction V (FrV) and Fraction IV-1 (Fr IV-1) intermediates. The desired Cohn Pool plasma volume and protein concentration were obtained without significant loss of various plasma derived proteins from different fractionations of Cohn Pool plasma.

[0124] PptG and Fr IV-1 were further purified at pilot scale to Immunoglobulin drug product and to A1PI drug product, respectively. FrV was purified at commercial scale to Albumin drug product.

[0125] Prior to Cohn fractionation process, about 4700 liters cryo-poor plasma were concentrated using an ultrafiltration membrane. For the plasma concentration step, pre-filtration (10 μm+0.5 μm) followed by batch TFF with UF cassettes with a nominal molecular weight cut off (NMWCO) of 100 kDa was employed. As a result, the starting protein concentrations of the Cohn Pool plasma in the test runs were about 15% higher than that in control run. For each protein of interest, technical specification of the separate and purified protein, for example, recovery rate, quality, are tabulated and analyzed.

1. Immunoglobulin Process Performance and Product Quality Were Not Affected by Cohn Pool Concentration

[0126] Immunoglobulin content was calculated following the concentration step. The resulting IgG content in the Cohn Pool plasma prepared for Ethanol treatment was concentrated without significant loss.

[0127] Proteins typically found in the Precipitate G and the Immunoglobulin drug product downstream from Cohn pool concentration are found comparable in yield and purity to those found in Precipitate G and Immunoglobulin drug product in a process starting with non-concentrated Cohn Pool.

[0128] All operational parameters for upstream and downstream processing such as cycle time and step yields were found comparable to historic commercial manufacturing data. IgG quality was evaluated at drug product level and all results met predefined limits as shown in Table 18.

TABLE-US-00029 TABLE 18 Immunoglobulin Drug Product Test Results Test Acceptance Criteria Units Commercial Scale Run Measles ≥0.22 times CBER Ref 0.54 Antibody Lot 176 Diphtheria ≥1.2 U of US standard U/mL 9.4 Antibody Antitoxin/mL Poliomyelitis ≥0.2 times the antibody 0.8 Antibody level of CBER Reference Polio Immune Globulin Lot 176 IgA ≤0.14 mg/mL 0.05 IgM ≤10 mg/dL <4 Anti-A NMT 1/64 @25 g/L of IgG 1/ 32 Anti-B NMT 1/64 @25 g/L of IgG 1/ 16 Anti-D NMT NIBSC Ref. — Satisfactory antibodies 02/228 (or equivalent) Anti- NMT 50 corresponding % 42 complement to 1 CH50 U/mg protein Activity Appearance The liquid preparation is — Satisfactory clear or slighlty opalescent and colorless or pale yellow Glycine 0.20-0.30/0.21-0.26 M 0.23 Purity ≥98 % 100 Osmolality 240-300 mOsmol/kg 259 pH 4.6-5.1

Diluted @ — 4.9 1% with 0.9% NaCl Prekallikrein ≤7 — <5 Activator Activity Ig G Subclass Similar to normal complies complies Distribution plasma Complement- ≥60 (cfr. EP) % 103 mediated Lysis of Red Blood Cells

3. Process Parameters for Immunoglobulin Separation and Purification From Cohn Pool in Commercial Scale Run

Cohn Pool was concentrated by about 15% (w/v) by ultrafiltration, forming a first Cohn Pool concentrate. The technical protocol and parameters associated with the process of ultrafiltration are indicated in Table 19.

TABLE-US-00030 TABLE 19 Parameters for Cohn Pool Concentration Step Description Unit Commercial Scale Run Cohn Pool mass before [kg] 4684 concentration Cohn Pool volume [L] 4565 before concentration Type of pre-filter N/A PP filters e.g., 10 μm + 0.5 μm or equivalent Pre-filter surface area [m.sup.2] 1.6 m.sup.2 (filter area selection dependent on desired filtration time) Pre-filtration time [hh:mm] 06:07 Type of filter for N/A OMEGA T Series concentration 100 kDa or equivalent Filter surface area [m.sup.2] 30 Number of filters used N/A 12 Filtration/concentration [hh:mm] 1:01 time Maximum sustained TMP [bar] 1.5 Average retentate flowrate [L/min] 8200 l/h Average permeate flowrate [g/min] approx. 700 L/H Initial concentration [%] 15% factor (by mass) Post-wash using 0.9% NaCl [L] 41 L (minimum amount to clear line) Cohn Pool mass after [kg] 3982.9 concentration & post-wash Cohn Pool volume after [L] 3882 concentration & post-wash Final concentration [%] 15% factor (by mass) Filter load [L CPP/m.sup.2] 152.2 Turbidity - before conc./ [NTU] 332/258 after conc.

[0129] In the exemplary CPC process, target amounts of filter aid and filter area were calculated based on the volume of concentrated Cohn Pool instead on the volume of cryo-poor plasma (CPP).

[0130] The first Cohn Pool concentrate was further submitted to fractionation procedure. The first Cohn Pool concentrate was contacted with 8% ethanol at a pH of from about 7.0 to 7.5 to obtain a Fraction I precipitate and a Fraction I supernatant from the fractionated Cohn Pool.

[0131] The Fraction I supernatant was further contacted with about 25% ethanol at a pH of from about 6.7 to about 7.3 to form a Fraction II+III precipitate.

[0132] The Fraction II+III precipitate was further suspended in a suspension buffer, thereby forming an IgG suspension. After mixing finely divided silicon dioxide (SiO.sub.2) with the IgG suspension for at least about 30 minutes, IgG suspension was filtered and resulted a filtrate and a filter cake.

[0133] The filtrate resulting from Fraction II+III precipitate was contacted with a detergent, forming a treated filtrate. After adjusting the pH of the treated filtrate to about 7.0, and adding ethanol to a final concentration of from about 20% to about 30%, a Precipitate G precipitate was formed.

3. Plasma Derived Products Isolated From Fraction IV and V of Fractioned Cohn Pool Were Not Affected by Cohn Pool Concertation

[0134] Proteins typically found in the Fraction IV and V downstream from Cohn pool concentration are found in these fractions in yields and purity comparable to those found in Fraction IV and V in a process starting with non-concentrated Cohn Pool.

[0135] The supernatant of Fraction II+III was further submitted to fractionation procedure. The supernatant of Fraction II+III was contacted with 25% ethanol to obtain Fraction IV-1 precipitate.

[0136] The supernatant of Fraction IV-1 was subsequently contacted with 40% ethanol to obtain Fraction IV-4 precipitate.

[0137] The supernatant of Fraction IV-4 was subsequently contacted with 40% ethanol to obtain Fraction V precipitate.

[0138] The quality of proteins isolated from Fraction V were evaluated through multiple biochemical and immunological assays as shown in Table 20. Quality indicating parameters of isolated proteins in Fraction V of the commercial scale run are comparable to historic commercial manufacturing data starting from non-concentrated Cohn Pool.

TABLE-US-00031 TABLE 20 Extended Characterization Testing at Fraction V Test Parameter Unit Commercial Scale Run PKA IU/mL <4 Purity % 97.2

[0139] Human Albumin and A1PI isolated from concentrated Cohn Pool plasma were tested at final drug product level. All results at drug product level are within predefined ranges as shown in Table 21-22.

[0140] All results met the predefined acceptance criteria at drug product level. Based on the illustrated results it can be concluded that Cohn Pool concentration does not negatively impact product quality of either A1PI or Albumin.

TABLE-US-00032 TABLE 21 20% Albumin Drug Product Test Results from Commercial Scale Run Test Acceptance Criteria Units Commercial Scale Run Visual Control clear, slightly viscous N/A Confirm solution; almost colourless, yellow to brown or green pH-Value 6.7-7.3 N/A 7.0 Potassium Content ≤2 μmol/mL <0.9 Aluminum Content ≤200 μg/L ≤50 μg/L Haem Content <0.15 Absorbance 0.05 unit Polymers and Area of the peak due to % Polymers + Aggregates (area %/ polymers and aggregates is Aggregates 6% protein %) not greater than 10 percent Polymers + of the total area of the Aggregates chromatogram/corresponding (proteins) 3% to not more than 5% of polymers and aggregates Prekallikrein ≤35 IU/mL IU/mL <4 Activator Activity Sodium Content ≤160 μmol/mL 123 Protein Composition ≥96% human albumin % human albumin 98.4% Protein Content 20% Albumin: 190-210 mg/mL 196 Heat Stability Test complies (remain Complies (remains Complies unchanged) unchanged) Bacterial Endotoxins 20%: ≤1.7 EU/mL <0.5 Test for Sterility sterile N/A sterile Particle ≥10 μm: ≤6000 particles/vial ≥10 μm: 744 Contamination ≥25 μm: ≤600 ≥25 μm: 24 Osmolality 210-400 mOsmol/kg 226

TABLE-US-00033 TABLE 22 A1PI Drug Product Test Results Commercial Test Acceptance Criteria Units Scale Run AAT Concentration 16-24 mg/mL mg/mL 21 AAT Specific Activity ≥0.35 mg AAT/mg mg AAT/mg 1.0 protein protein Appearance/Visual Solution is clear N/A satisfactory Inspection and colorless to yellow-green May contain a few protein particles pH 6.5-7.8 — 7.0

[0141] In additional lab scale filtration experiments it was demonstrated that permeate flux was comparable for filters up to 300 kDa molecular weight cut-off when concentrating Cohn Pool.

Claims

1. A method for preparing a Cohn Pool concentrate from blood plasma, the method comprising: (a) providing a Cohn Pool from blood plasma; and (b) concentrating the Cohn Pool to a total protein concentration of at least about 65 g/L, thereby forming a Cohn Pool concentrate.

2. The method of claim 1, wherein the Cohn Pool is concentrated to a protein concentration of at from about 50 to about 65 g/L

3. The method of any preceding claim, wherein the Cohn Pool is a member selected from cryo poor plasma, Factor poor plasma, and Inhibitor poor plasma.

4. The method of any preceding claim, wherein step (b) is performed using an ultrafiltration membrane with a nominal molecular weight cut off (NMWCO) of 300 kDa or less run either in re-circulation or in single-pass configuration, either with or without preceding pre-filtration.

5. The method of any preceding claim, wherein step (b) is performed using a hollow-fiber filter membrane with a nominal molecular weight cut off (NMWCO) of 300 kDa or less run either in re-circulation or in single-pass configuration, either with or without preceding pre-filtration.

6. The method of any preceding claim, wherein in the Cohn Pool concentrate has a total protein concentration of at least about 65 g/L.

7. The method of any preceding claim, wherein the Cohn Pool concentrate is further subjected to a purification process for preparing a composition selected from an immunoglobulin G (IgG) composition, an albumin composition, an A1-PI composition and a combination thereof.

8. The method of any preceding claim, wherein Cohn Pool is concentrated with essentially no loss of protein mass attributable to a member selected from Total Protein, IgG, albumin, AAT, and fibrinogen in the resulting Cohn Pool concentrate.

9. The method of any preceding claim, further comprising submitting the Cohn Pool concentrate to a plasma fractionation procedure.

10. The method of claim 9, wherein the fractionation procedure is Cohn Fractionation or one of its modifications.

11. The method of claim 10, wherein the fractionation procedure comprises: (i) contacting the Cohn Pool concentrate with from about 6% to about 10% ethanol at a pH of from about 7.0 to 7.5 to obtain a Fraction I precipitate and a Fraction I supernatant; and (ii) contacting the Fraction I supernatant or Cohn pool concentrated with from about 18% to about 27% alcohol at a pH of from about 6.7 to about 7.3 to form a member selected from a Fraction II+III precipitate or alternatively a Fraction I+II+III precipitate.

12. The method of claim 11, further comprising: (iii) suspending a member selected from the Fraction II+III precipitate, the Fraction II+III (II+III or alternatively I+II+III) precipitate in a suspension buffer, thereby forming an IgG suspension; (iv) mixing finely divided silicon dioxide (SiO.sub.2) with the IgG suspension for at least about 30 minutes; (v) filtering the IgG suspension, thereby forming a filtrate and a filter cake.

13. The method of claim 12, further comprising: (vi) contacting the filtrate with a detergent, forming a treated filtrate; (vii) adjusting the pH of the treated filtrate of step (vi) to about 7.0, and adding ethanol to a final concentration of from about 20% to about 30%, thereby forming a Precipitate G precipitate; (viii) dissolving the Precipitate G precipitate in an aqueous solution comprising a member selected from a solvent a detergent and a combination thereof, forming a Precipitate G solution; (ix) passing the solution through a cation exchange material, adsorbing proteins contained therein onto the cation exchange material, and subsequently eluting the adsorbed proteins in an eluate; (x) passing the eluate through an anion exchange material generating a flow-through effluent; (xi) passing the effluent through a nanofilter, generating a nanofiltrate; (xii) concentrating the nanofiltrate by ultrafiltration, generating a first ultrafiltrate; (xiii) diafiltering the first ultrafiltrate against a diafiltration buffer, generating a diafiltrate; and (xi) concentrating the diafiltrate by ultrafiltration, generating a second ultrafiltrate having a protein concentration between about 8% (w/v) and about 22% (w/v), thereby forming an IgG enriched fraction.

14. The method of claim 13, further comprising, prior to (vi), washing the filter cake with at least 1 filter press dead volume of a wash buffer having a pH of from about 4.9 to about 5.3, thereby forming a wash solution; combining the filtrate with the wash solution, thereby forming a solution, and treating the solution with a detergent in step (vi).

15. A plasma protein preparation, wherein the protein is a member selected from IgG, A1PI, and albumin prepared by the method of claim 12.

16. A pharmaceutical formulation comprising albumin isolated from the Cohn Pool concentrate of claim 1, and a pharmaceutically acceptable vehicle.

17. A pharmaceutical formulation comprising AAT isolated from the Cohn Pool concentrate of claim 1, and a pharmaceutically acceptable vehicle.

18. A pharmaceutical formulation comprising IgG isolated from the Cohn Pool concentrate of claim 1, and a pharmaceutically acceptable vehicle.

19. A pharmaceutical formulation comprising fibrinogen isolated from the Cohn Pool concentrate of claim 1, and a pharmaceutically acceptable vehicle.

20. A pharmaceutical formulation comprising TP isolated from the Cohn Pool concentrate of claim 1, and a pharmaceutically acceptable vehicle.
